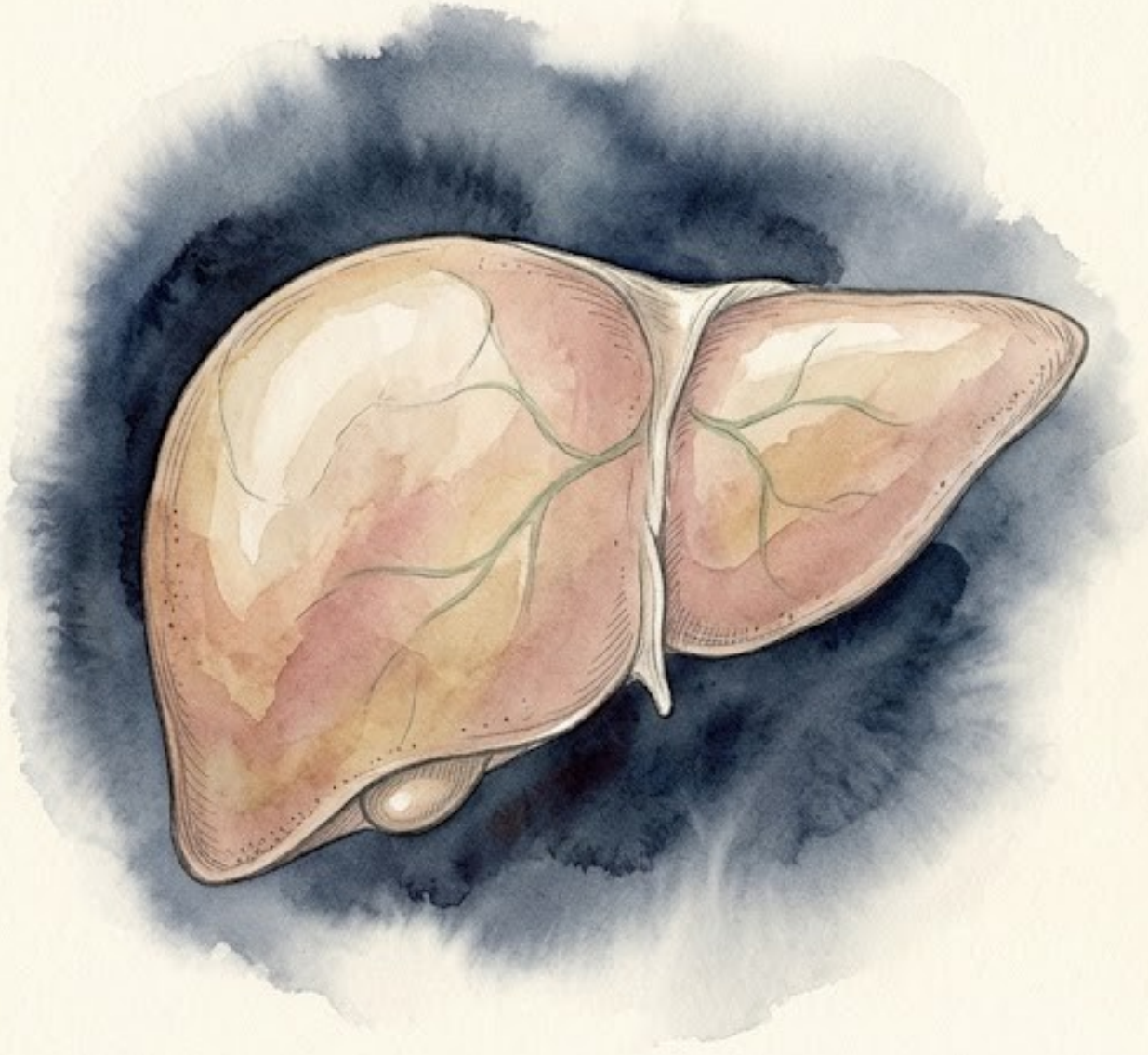


# Acute Liver Failure and ACLF in the ICU

**Mohammad Almeqdadi, MD**
Transplant Hepatologist and Gastroenterologist
Assistant Professor of Medicine and Surgery
X: @mfalmeqdadi

Tufts University | School of Medicine

Slides (PDF)

*Illustrations generated with Google Gemini*

# Objectives

*By the end of this talk you should be able to:*

- Tell acute liver failure apart from severe acute liver injury and from ACLF
- Start the first-hour care of ALF: cause, NAC, glucose, airway
- Protect the brain in ALF and know when to intubate
- Start CRRT for ammonia without waiting for a renal indication
- Diagnose and grade ACLF, and find the trigger
- Work through AKI in cirrhosis, including HRS-AKI, without hurting the lungs
- Know when to call transplant hepatology

# Two patients, same ICU

| Bed 4 | Bed 7 |
|---|---|
| 28-year-old woman | 56-year-old man |
| No known liver disease | Alcohol-related cirrhosis with ascites |
| Nausea and vomiting for 4 days | Fever, confusion, low blood pressure |

*Both have a sick liver. Do they have the same disease?*

# Bed 4

28F, acute liver failure



# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Alkaline phosphatase | 128 U/L |
| Total bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

**Is this acute liver failure?**

# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Alkaline phosphatase | 128 U/L |
| Total bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

**Is this acute liver failure?**

**We don't know yet. What's missing?**

# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Alkaline phosphatase | 128 U/L |
| Total bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

## What's missing?

- **INR** (synthetic function, not enzymes)
- **Mental status** (any encephalopathy?)
- **Time course** (how long since the first symptom?)
- **Is there chronic liver disease underneath?**

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Total bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

**Now is it acute liver failure?**

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Total bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

**This is acute liver failure**

# Acute liver failure

- **INR ≥ 1.5** and **any degree of encephalopathy**
- Illness of less than 26 weeks
- No pre-existing cirrhosis
  - AASLD and ACG: chronic liver disease without cirrhosis still allows ALF. EASL: no prior liver disease at all
  - Wilson disease, autoimmune hepatitis and vertically acquired HBV can still count if recognized within 26 weeks

| **Jaundice to encephalopathy** | **Hyperacute (< 1 week)** | **Acute (1 to 4 weeks)** | **Subacute (4 to 12 weeks)** |
|---|---|---|---|
| Typical causes | Acetaminophen, HAV, HEV, ischemia | HBV | Idiosyncratic drug injury, AIH, indeterminate |
| Cerebral edema | **Common** | Less common | Uncommon |
| Recovery without transplant | More likely | Intermediate | **Less likely** |

*EASL ALF guideline 2017 [PMID: 28417882]; ACG ALF 2023 [PMID: 37377263]; Management of ALF: update 2022 [PMID: 36001996]*

# INR 2.4, wide awake. Acute liver failure?

**What do you call this, and what do you do now?**

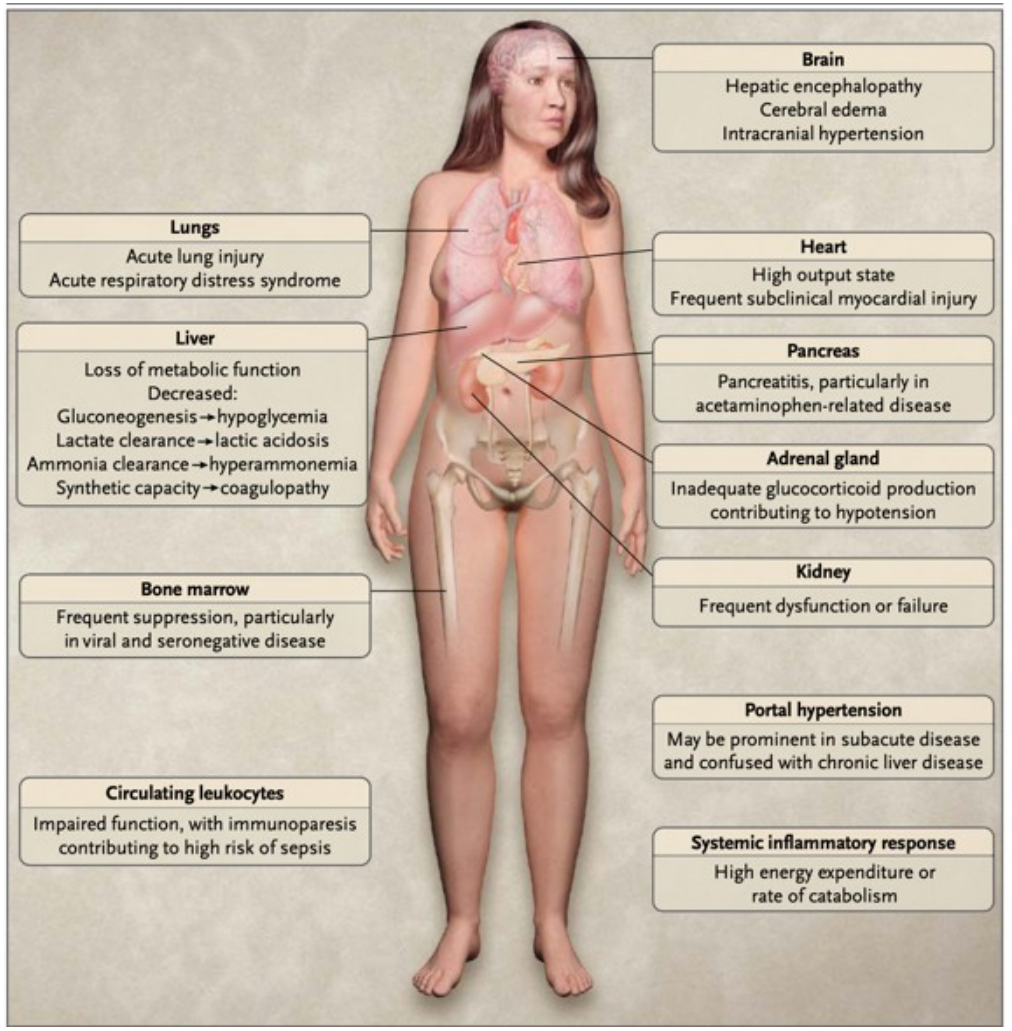
# INR 2.4, wide awake. Acute liver failure?

- **No.** This is severe acute liver injury
- No encephalopathy means not yet ALF
- But it can become ALF within hours
  - Check glucose often and examine the patient every few hours
  - Call transplant hepatology early anyway
- And the opposite trap:
  - Hypoglycemia, sedatives, sepsis, alcohol withdrawal and seizures can all look like encephalopathy

# What else fails?

**It's called liver failure. Which other organs fail?**

# What else fails?




*Bernal W, Wendon J. N Engl J Med 2013;369:2525-34, Figure 1*

28F, 4 days of nausea

**What's your next step?**

# Three things, at the same time

| Keep her alive | Find and treat the cause | Call transplant hepatology |
|---|---|---|
| ICU bed<br>Glucose every 1 to 2 hours<br>Airway assessment<br>Volume, perfusion, lactate<br>Neuro checks | Acetaminophen level<br>Viral, autoimmune, Wilson, toxicology<br>Doppler ultrasound<br>Pregnancy test<br>Start NAC if acetaminophen is possible | Now, not after the workup<br>Transplant evaluation starts on admission<br>Social work and psychiatry while she can still talk |

**All three start in the first hour**

# Which cause changes treatment right away?

| If you suspect | Send | Don't wait for the result to |
|---|---|---|
| Acetaminophen | Level, ingestion history, combination products | |
| Herpes simplex | HSV PCR (blood) | |
| Hepatitis B | HBsAg, anti-HBc IgM, HBV DNA | |
| Autoimmune hepatitis | ANA, ASMA, IgG (can be negative) | |
| Wilson disease | Ceruloplasmin, urine copper, hemolysis, low ALP | |
| Pregnancy (AFLP, HELLP) | hCG | |
| Budd-Chiari | Doppler ultrasound or CT | |
| Ischemic hepatitis | Echo, history of shock | |
| Mushrooms (Amanita) | Wild mushrooms, then diarrhea 6 to 24 hours later | |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

# Which cause changes treatment right away?

| If you suspect | Send | Don't wait for the result to |
|---|---|---|
| Acetaminophen | Level, ingestion history, combination products | Start NAC |
| Herpes simplex | HSV PCR (blood) | Start IV acyclovir |
| Hepatitis B | HBsAg, anti-HBc IgM, HBV DNA | Start entecavir or tenofovir |
| Autoimmune hepatitis | ANA, ASMA, IgG (can be negative) | Discuss steroids with hepatology, list in parallel |
| Wilson disease | Ceruloplasmin, urine copper, hemolysis, low ALP | Prepare for urgent transplant |
| Pregnancy (AFLP, HELLP) | hCG | Call obstetrics, plan delivery |
| Budd-Chiari | Doppler ultrasound or CT | Anticoagulation, discuss TIPS or transplant |
| Ischemic hepatitis | Echo, history of shock | Fix the circulation |
| Mushrooms (Amanita) | Wild mushrooms, then diarrhea 6 to 24 hours later | NAC, call poison control (silibinin if available) |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

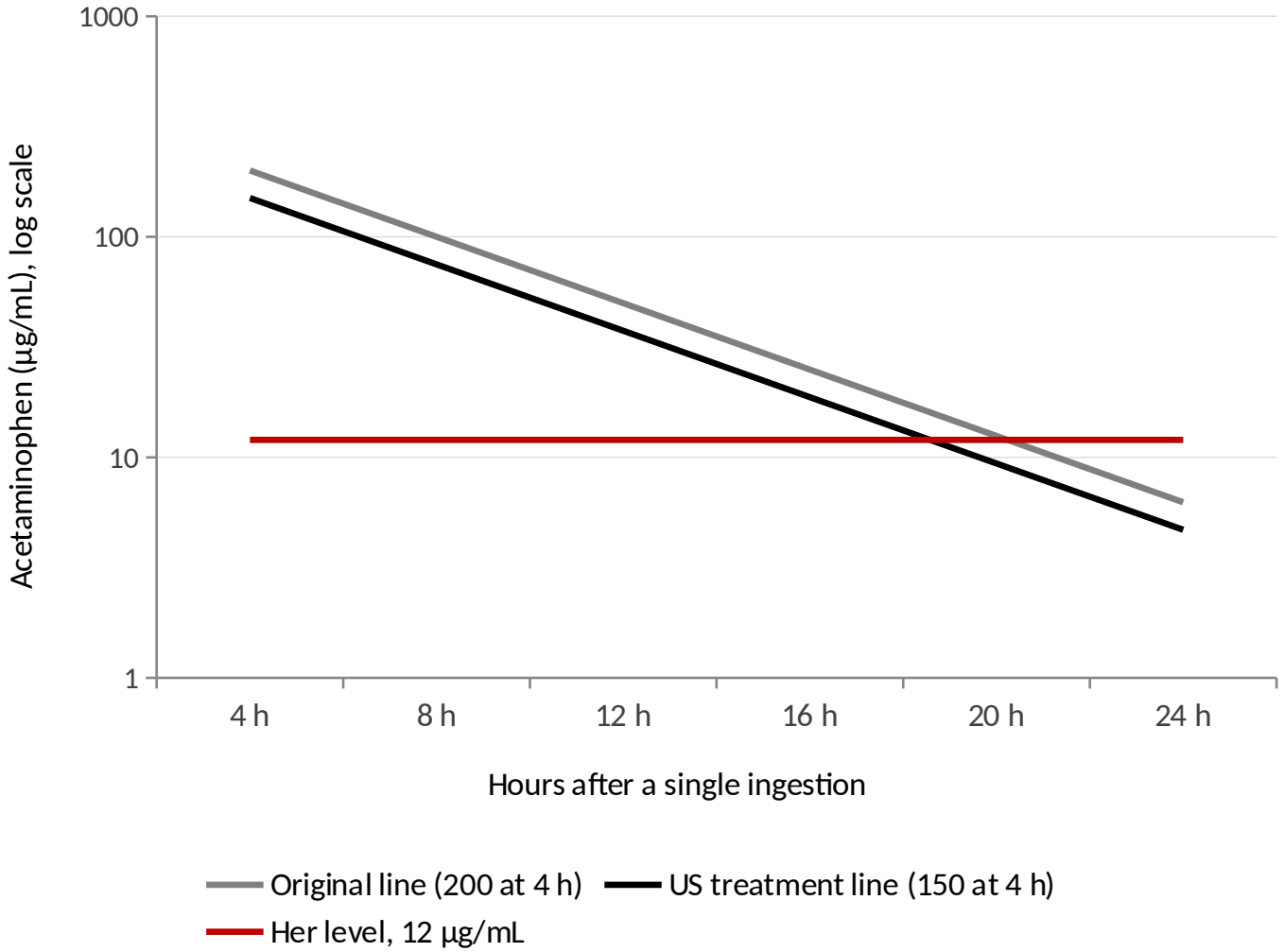
# 28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown




**Can you use the nomogram?**

# The Rumack-Matthew nomogram




- **One acute ingestion at a known time: it tells you when NOT to give NAC**
- Late, repeated or unknown-time ingestion: a low level doesn't rule out liver injury

**If she had taken it all at once**

- 8 hours: 12 is below the line, no NAC
- 20 hours: above the line, NAC
- But it was 3 days of repeated doses. **There is no time zero**. Give NAC

*Rumack BH, Matthew H. Pediatrics 1975; US/Canada acetaminophen consensus 2023 [PMID: 37552484]*

# 28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown




## Can you use the nomogram?

- **No.** Rumack-Matthew is for a single acute ingestion with a known time
- Repeated supratherapeutic use and late presentation: the level can be low or undetectable
- **Start NAC now**

# NAC for every kind of acute liver failure?

Published in final edited form as:
*Gastroenterology*. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

## INTRAVENOUS *N*-ACETYLCYSTEINE IMPROVES TRANSPLANT-FREE SURVIVAL IN EARLY STAGE NON-ACETAMINOPHEN ACUTE LIVER FAILURE

**WM Lee**[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

**What do you think they found?**

# NAC for every kind of acute liver failure?

Published in final edited form as:
*Gastroenterology*. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

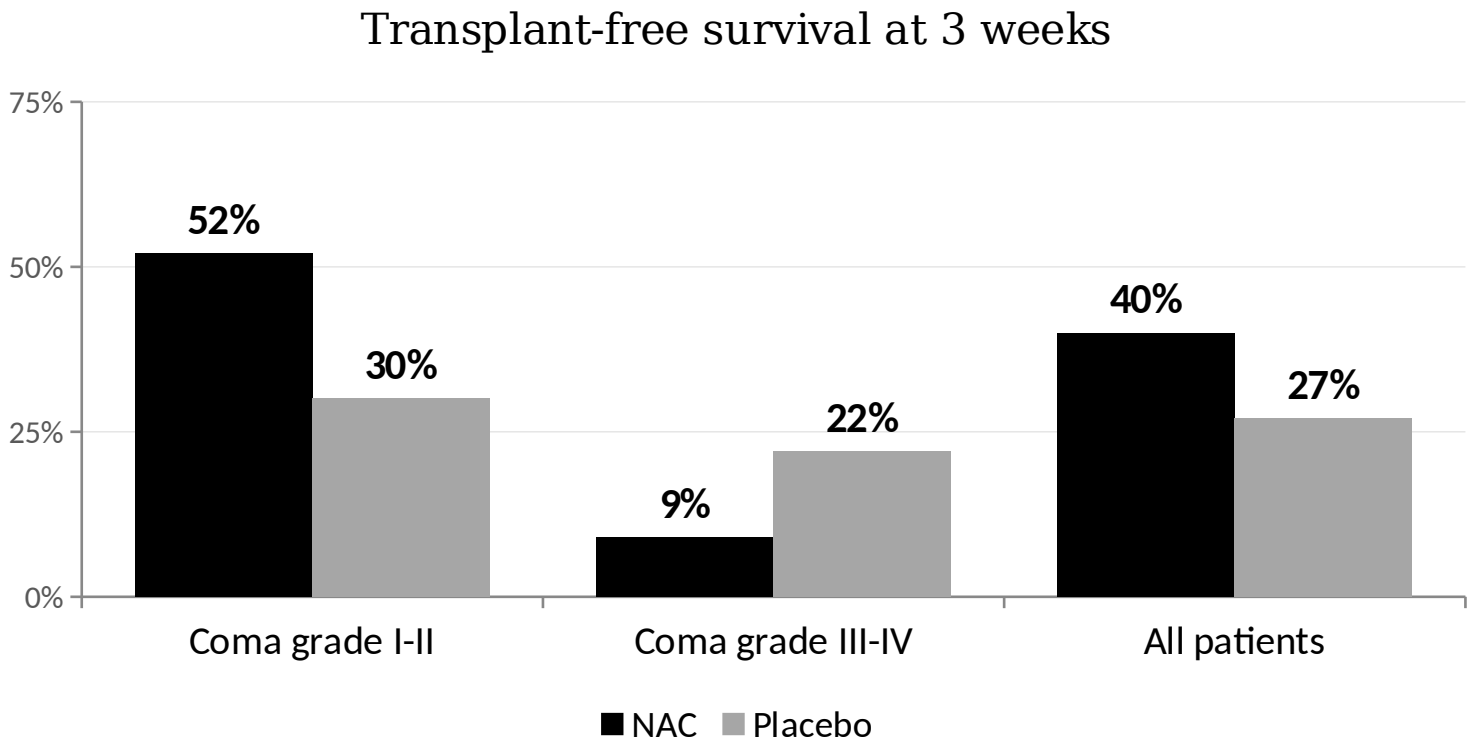
## INTRAVENOUS *N*-ACETYLCYSTEINE IMPROVES TRANSPLANT-FREE SURVIVAL IN EARLY STAGE NON-ACETAMINOPHEN ACUTE LIVER FAILURE

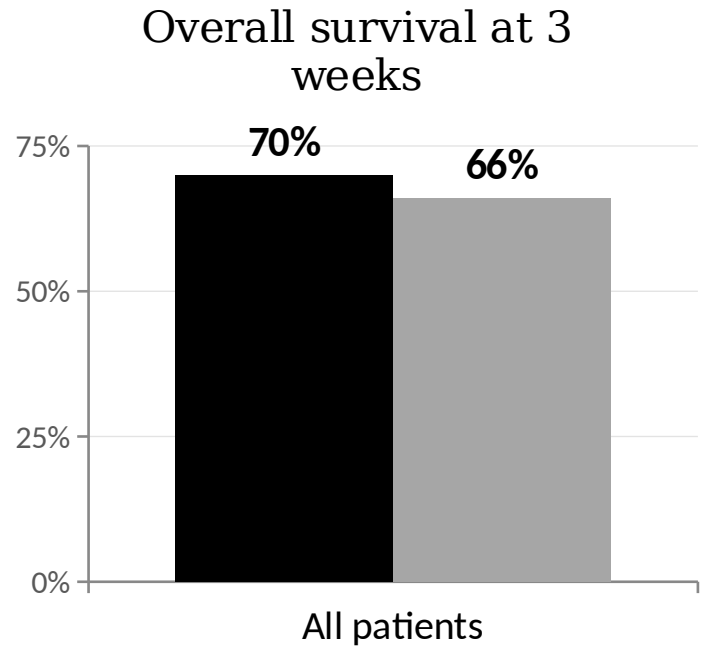
**WM Lee**[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

**Conclusions—**Intravenous NAC improves transplant-free survival in patients with early stage non-acetaminophen related acute liver failure. Patients with advanced coma grades do not benefit from NAC and typically require emergency liver transplantation. (ClinicalTrials.gov number NCT00004467)

# NAC for every kind of acute liver failure?

*Adult non-acetaminophen ALF, randomized, NAC vs placebo, 173 patients*



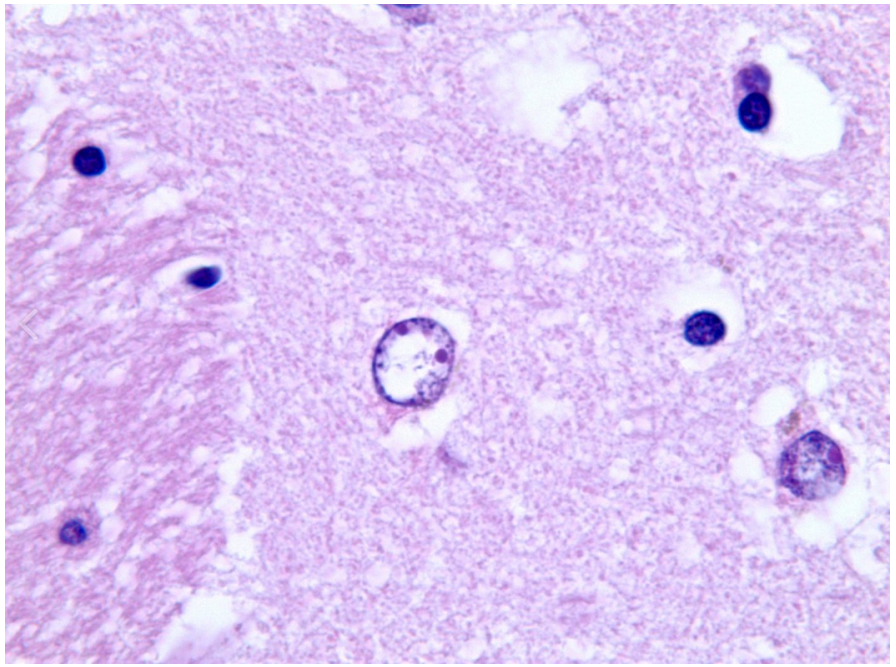




**Reasonable in early non-acetaminophen ALF. Don't extrapolate to advanced coma, children or ACLF**

*Lee WM et al. Gastroenterology 2009 [PMID: 19524577]*

# Her brain

Hour 12



*Alzheimer type II astrocytes: the brain adapting over months of cirrhosis. In ALF there are no months*

# 28F, 4 days of nausea

| | Admission | Hour 12 |
|---|---|---|
| Mental status | Disoriented, asterixis | **Responds only to voice (grade III)** |
| Ammonia (arterial) | Not sent | **182 µmol/L** |
| Urine output | OK | 15 mL/h for 4 hours |
| Lactate (after 3 L) | 2.1 mmol/L | 4.6 mmol/L |
| Creatinine | 1.0 mg/dL | 1.3 mg/dL |
| Sodium | 136 mmol/L | 133 mmol/L |
| Glucose | 54 mg/dL | Needs D10 |

## What worries you most?

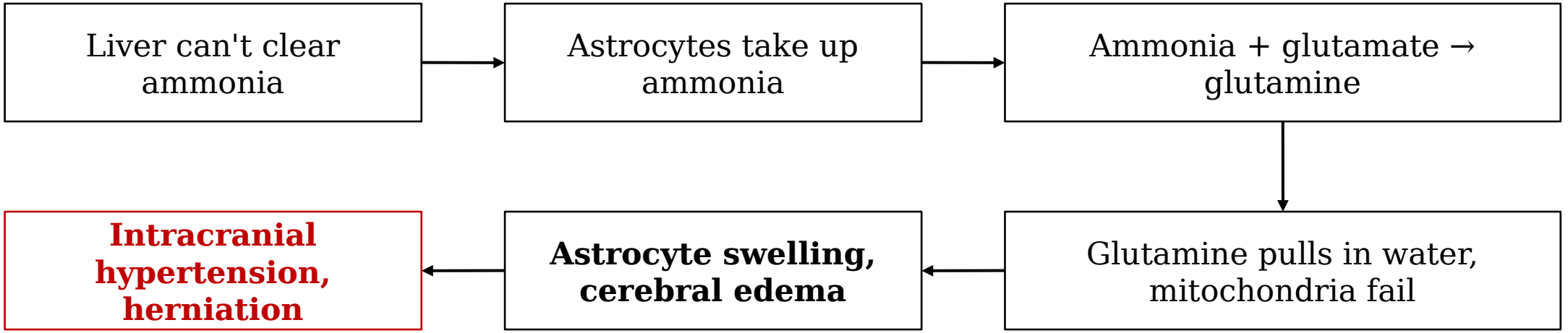
# 28F, 4 days of nausea

| | Admission | Hour 12 |
|---|---|---|
| Mental status | Disoriented, asterixis | **Responds only to voice (grade III)** |
| Ammonia (arterial) | Not sent | **182 µmol/L** |
| Urine output | OK | 15 mL/h for 4 hours |
| Lactate (after 3 L) | 2.1 mmol/L | 4.6 mmol/L |
| Creatinine | 1.0 mg/dL | 1.3 mg/dL |
| Sodium | 136 mmol/L | 133 mmol/L |
| Glucose | 54 mg/dL | Needs D10 |

**Her brain**

*When do you intubate? Do you wait for the creatinine before dialysis?*

# Why does the brain swell?



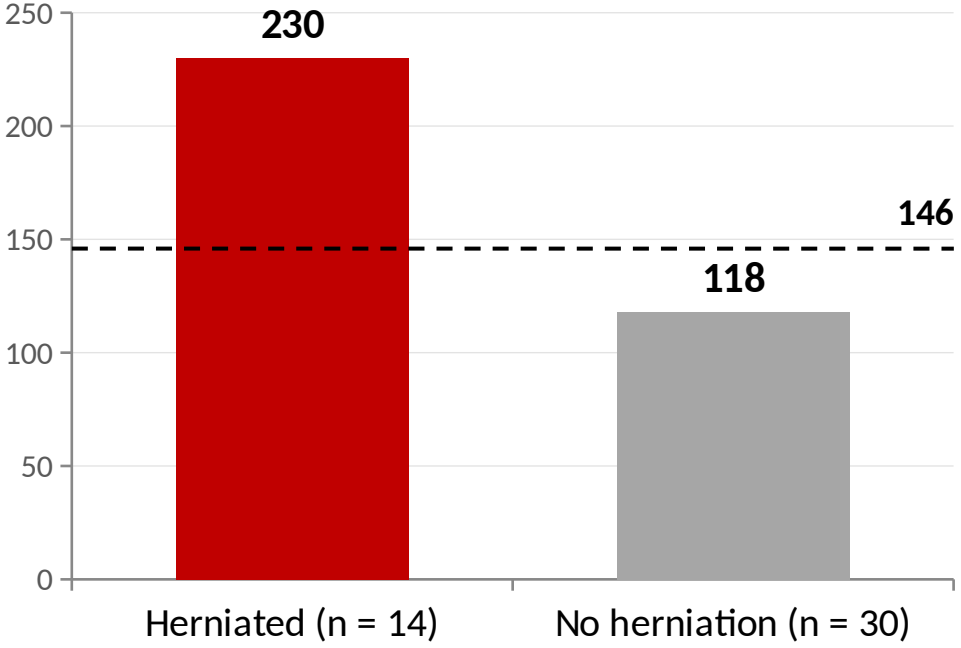

- **Makes it worse:** inflammation and infection, loss of autoregulation, fever, low sodium, high CO2
- **Highest risk:** hyperacute course, young, grade III-IV encephalopathy, ammonia persistently above 150 to 200 µmol/L, AKI, vasopressors

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]*

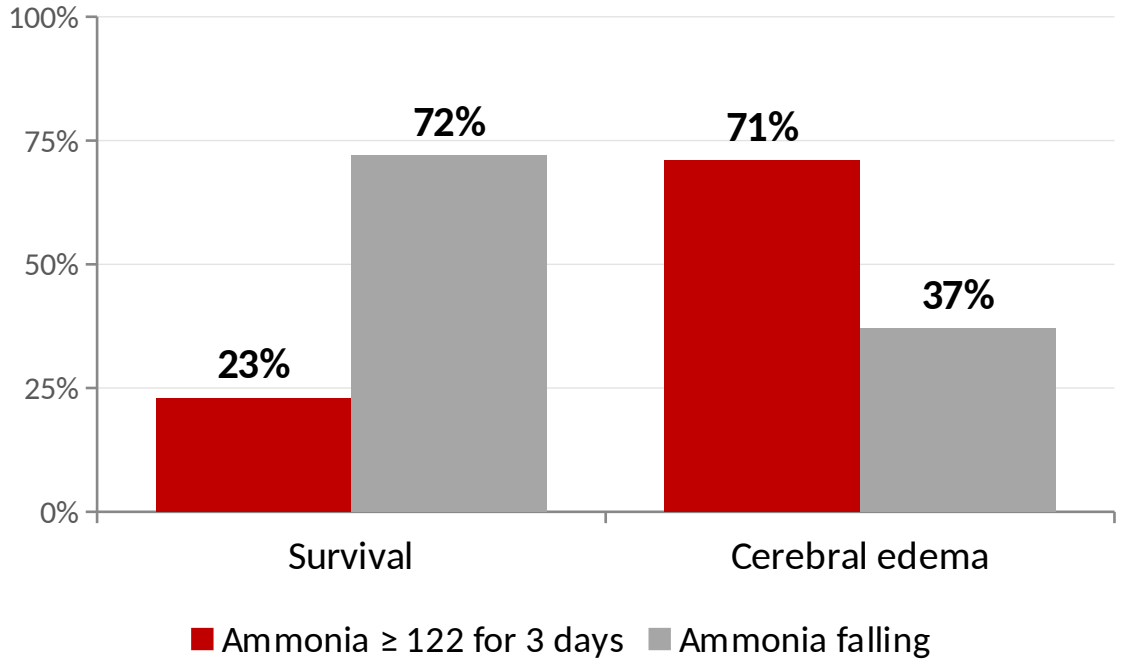



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients



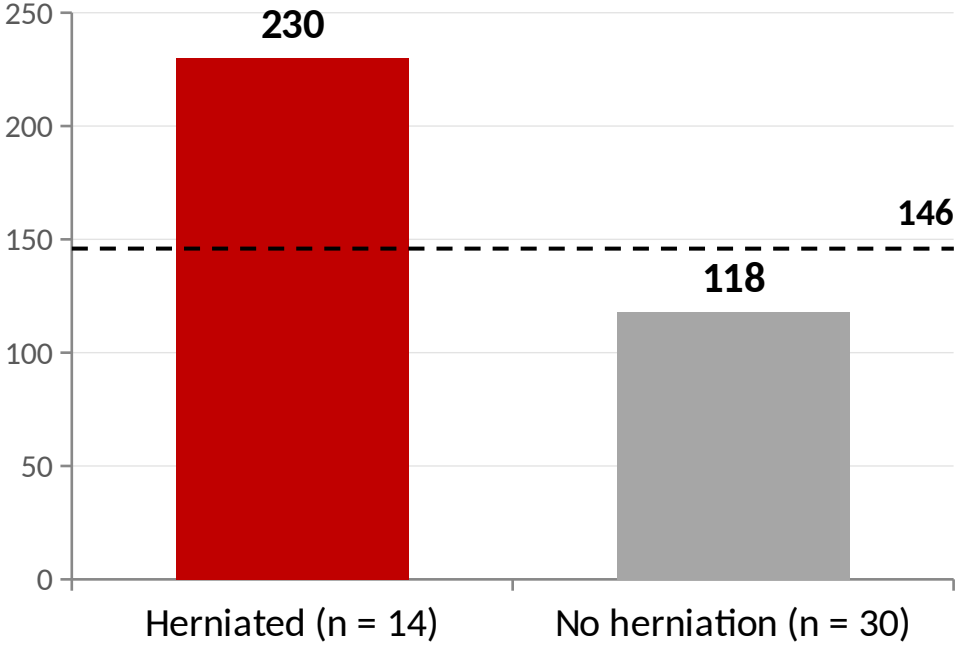

**So what number worries you?**

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

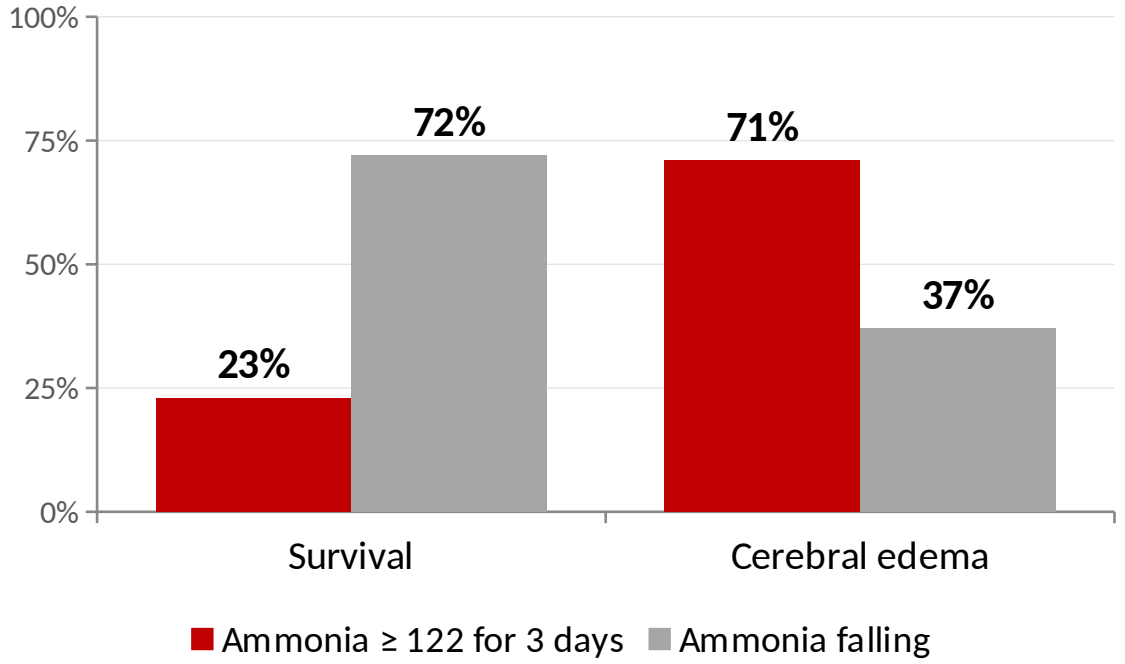



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients




**Above 150 to 200 and not falling: worry about the brain**

*No single level rules ICH in or out. Read it with the HE grade and the trend*

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

**Yes or no?**

# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

| Why people give it | Why not |
|---|---|
| Ammonia drives the brain swelling | **No proven benefit. ALFSG matched comparison: a little longer survival time, no difference in encephalopathy or outcome** |
| It's what we do in cirrhosis | Gas-distended bowel gets in the surgeon's way at transplant |
| Cheap, and the AASLD position paper allows it in early grades, short of diarrhea | Diarrhea means free water loss and volume swings, which fight your sodium target |
| | Liquid by mouth or tube in a patient who can't protect her airway: aspiration |

**Not for her: CRRT for the ammonia, protect the airway, keep the sodium up**

*AASLD position paper 2011, rec. 26; Alba L, Hay JE, Lee WM et al. J Hepatol 2002;36(Suppl 1):33 (abstract); ACG ALF 2023 [PMID: 37377263]*

# When do you intubate?

- Responds to voice, drifts off, intermittently agitated
- Handling secretions for now
- SpO2 96% on 2 L

**Intubate now, or wait until she can't protect her airway?**

# When do you intubate?

## Grade III-IV encephalopathy in ALF: intubate

| Before | During | After |
|---|---|---|
| Preoxygenate<br>Norepinephrine ready<br>Check glucose<br>Head of bed 30°, neck neutral | Avoid hypotension (perfusion pressure)<br>Avoid coughing and bucking (ICP spikes)<br>Most experienced operator | Propofol: short-acting, may lower ICP<br>Avoid long-acting benzodiazepines<br>Aim for normal CO2, check a gas<br>Sedation hides the neuro exam: check pupils |

*Practical update 2024 [PMID: 39170946]; ACG ALF 2023 [PMID: 37377263]; SCCM 2020 [PMID: 32058387]*

# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

**Lung protection or brain protection?**

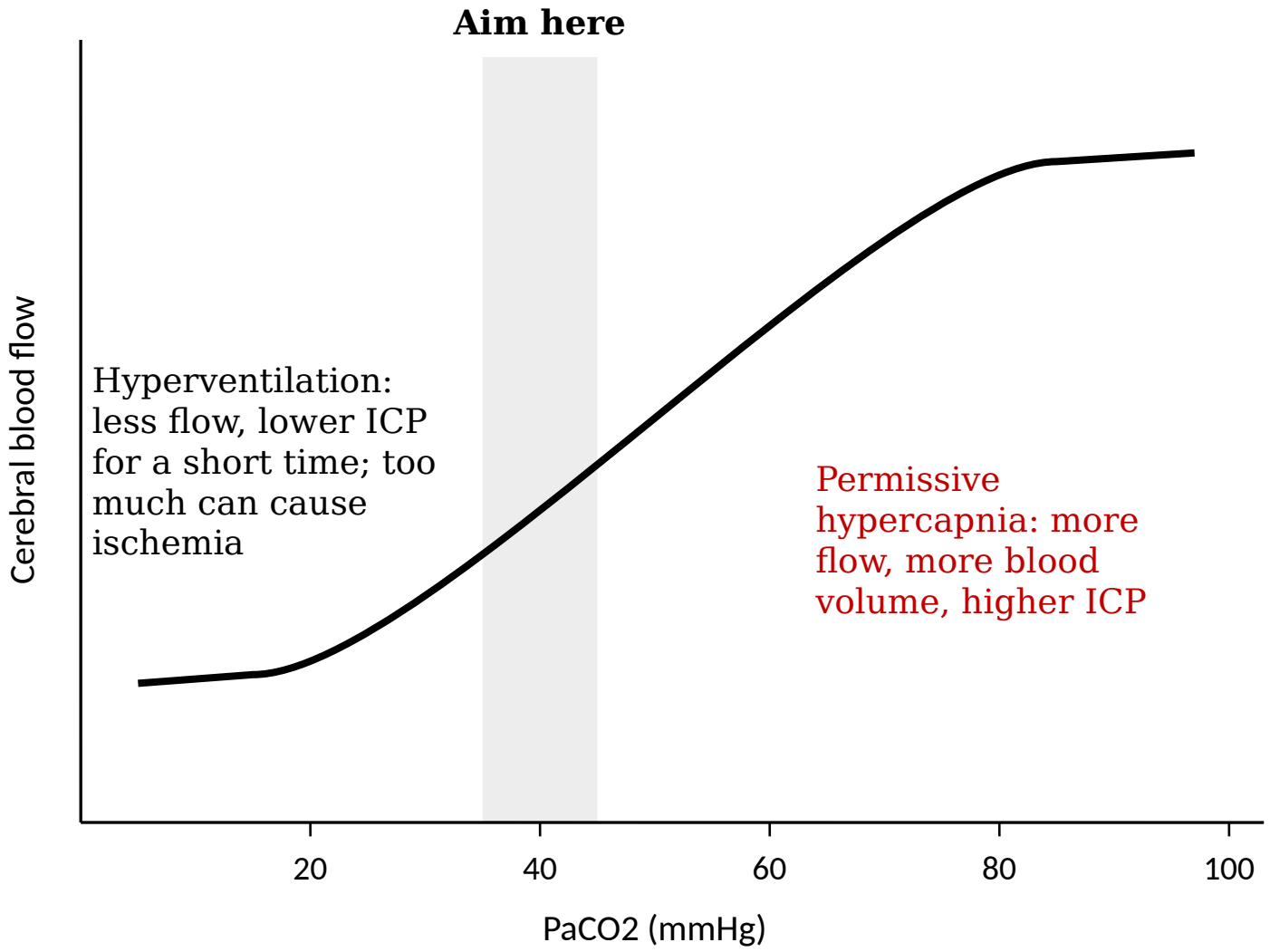
# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

| | The lung wants | The brain wants | What we do |
|---|---|---|---|
| **Tidal volume** | 6 mL/kg predicted body weight | Fine, but low volumes can push the CO2 up | Do it, and raise the rate to hold the CO2 |
| **CO2** | Tolerate a high CO2 | High CO2 dilates vessels and raises ICP | **Aim for normal CO2. No permissive hypercapnia when ICP is a concern** |
| **Hyperventilation** | No | Lowers ICP briefly; if overdone or prolonged, can cause ischemia | Only as a short rescue for herniation |
| **PEEP** | Recruit | Very high pressures can impair venous return | Set it for oxygen and blood pressure. PEEP is not forbidden |

*SCCM 2020 [PMID: 32058387]; practical update 2024 [PMID: 39170946]*

# CO2 and cerebral blood flow

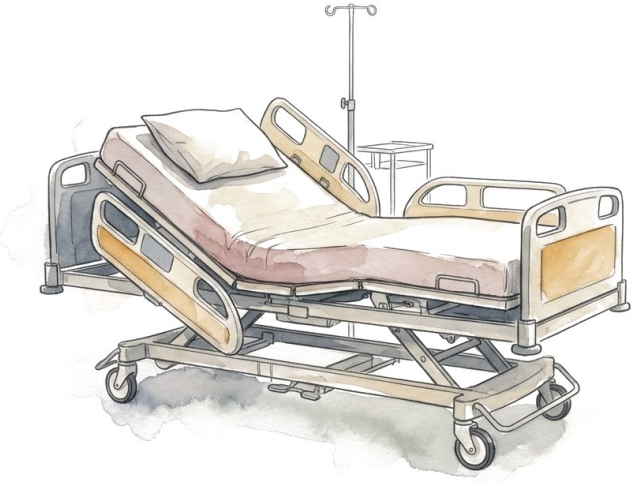



- Between about 20 and 80 mmHg, cerebral blood flow rises with PaCO2
- In a swollen brain, more blood volume means a higher ICP
- Keep PaCO2 normal. Hyperventilate only as a short bridge for herniation

*Schematic, not to scale. Standard cerebral physiology*

# Protecting the brain: every patient with grade III-IV

- Head of bed 30°, neck neutral
- Treat fever, avoid hypoglycemia
- Sodium **145 to 155 mmol/L** with hypertonic saline (use your center's protocol)
- Avoid fluid overload
- Treat seizures, EEG if in doubt. No routine prophylaxis
- No cooling: 34 °C did not prevent intracranial hypertension in an RCT



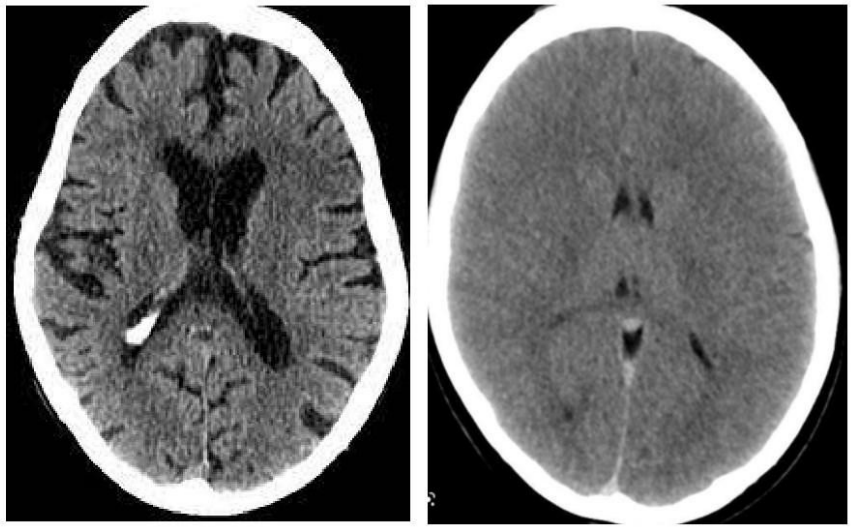
*SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]; Bernal W et al. J Hepatol 2016;65:273-9*

# A surge: new hypertension, bradycardia or a pupil

- Hyperosmolar bolus (hypertonic saline or mannitol)
- Brief hyperventilation as a bridge
- Call the transplant team now

**Things to know**

- Hypertension and bradycardia come late. Don't wait for them
- A normal head CT does not exclude raised ICP
- ICP monitors are not routine; SCCM suggests against (very low certainty)



Normal | Effacement of sulci

*SCCM 2023 [PMID: 37052436]; practical update 2024 [PMID: 39170946]*

# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

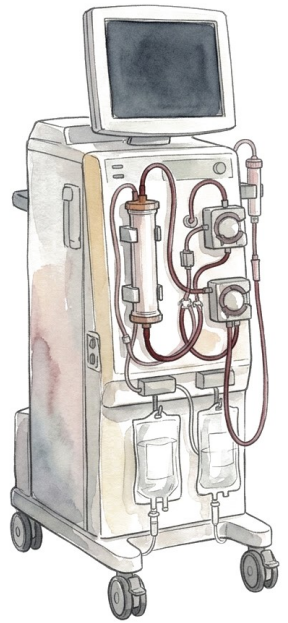
**Do you wait for a conventional dialysis indication?**

# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

**No. In ALF, start CRRT for the ammonia**

- ACG 2023: suggests early CRRT for hyperammonemia with grade II or higher encephalopathy, even without a renal indication (conditional, very low certainty)
- Continuous, not intermittent: fewer swings in blood pressure and brain water
- Call nephrology now



*ACG ALF guideline 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]*

# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

**2 units of FFP first?**

# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

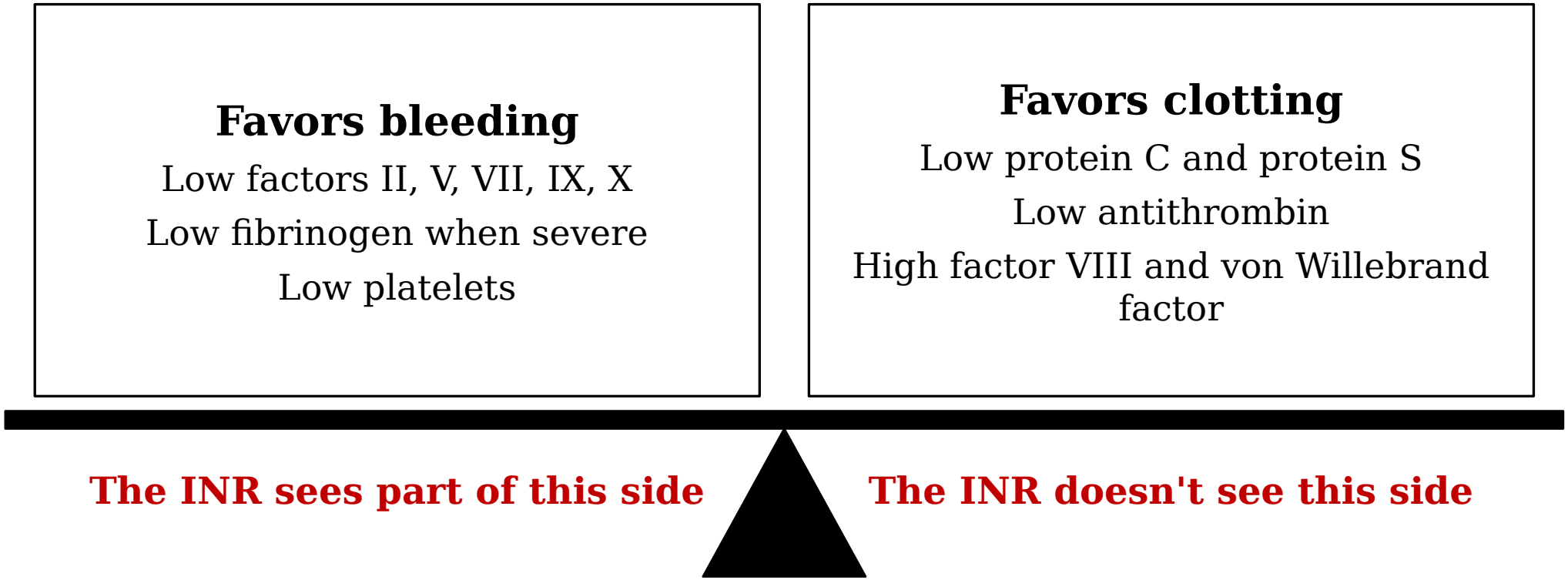
- **No plasma just to fix the number**
- The INR sees only the procoagulant side. Protein C and S are low too, factor VIII and von Willebrand factor are high. Most patients are rebalanced
- Plasma adds volume (lungs, brain) and erases the INR trend that prognosis and listing rely on

**What to do instead**

- Ultrasound-guided line, experienced operator, compressible site
- Vitamin K once; check fibrinogen and platelets; viscoelastic testing if available
- Treat bleeding. Plan high-risk procedures (like an ICP monitor) separately

*Update 2022 [PMID: 36001996]; SCCM 2020 [PMID: 32058387]*

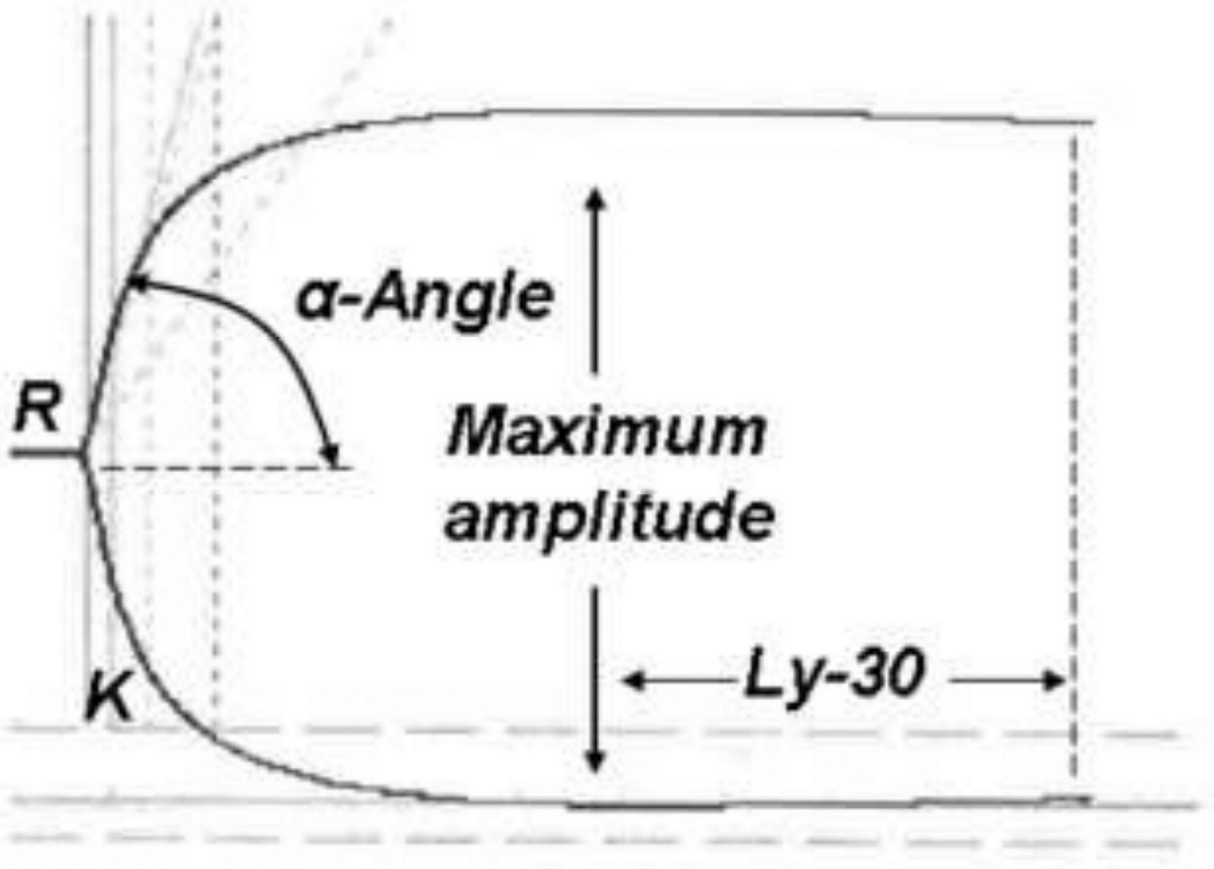
# Rebalanced hemostasis




*Balanced in most patients, but easily tipped either way: both bleeding and thrombosis happen*

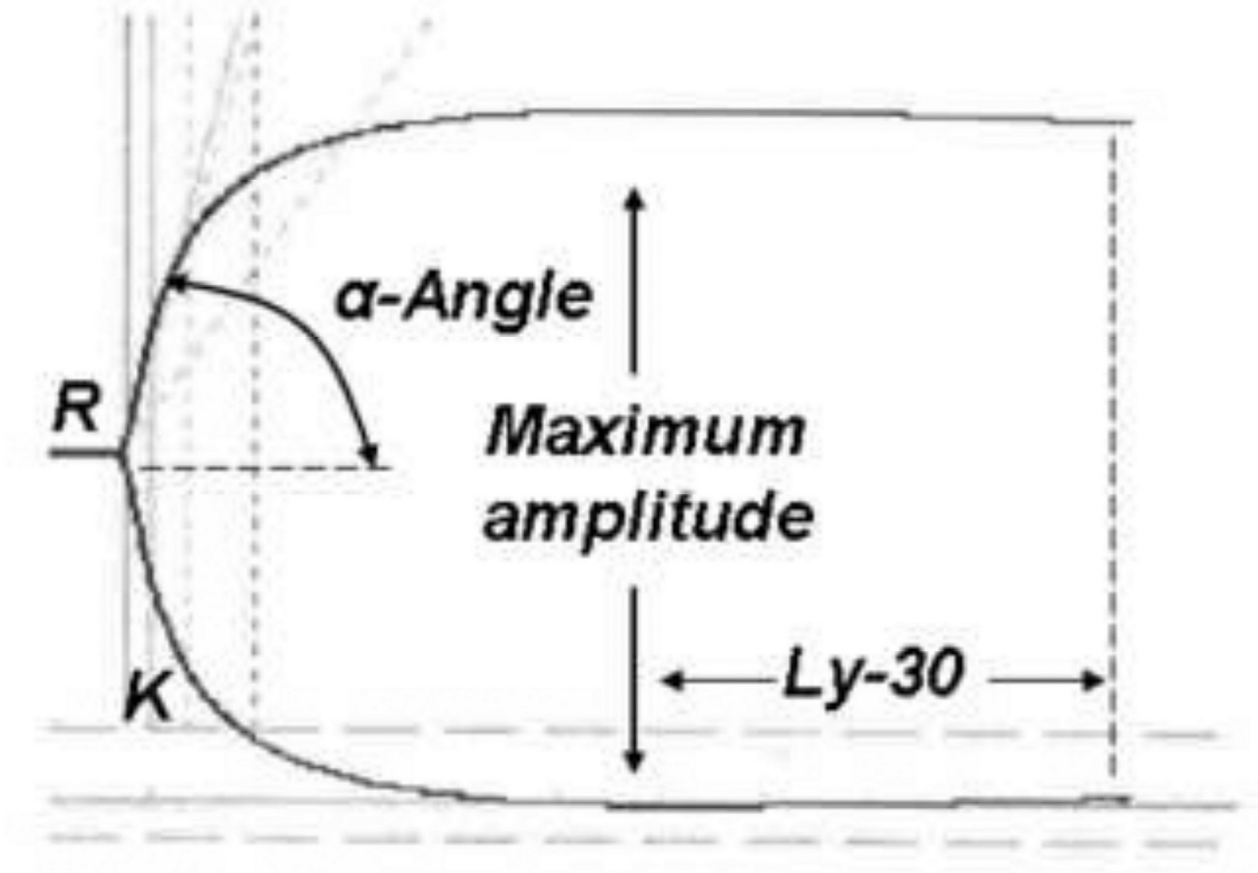
*Concept from Lisman T, Porte RJ. Blood 2010*

# INR 4.2. What does the TEG say?




*Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]*

# INR 4.2. What does the TEG say?




*Normal in all five TEG measures, in a patient with ALF and an INR of 4.2*

*Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]*

# CRRT for ammonia: what to agree on with nephrology

| | What to watch and say |
|---|---|
| **Why** | It's for the ammonia and the brain, not the creatinine. Give them the HE grade, the ammonia trend, urine output and fluid status |
| **Dose** | Prescribed vs delivered. Filter clotting and downtime mean less clearance, and the ammonia climbs back. Recheck it |
| **Sodium** | One plan for replacement fluid or dialysate, hypertonic saline and every other sodium source. Check the sodium often |
| **Safety** | Phosphate and potassium fall. Blood pressure drops at connection. Anticoagulation per your protocol; citrate can build up in liver failure, so follow the total to ionized calcium ratio |

*ACG ALF 2023 [PMID: 37377263]; Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; update 2022 [PMID: 36001996]*

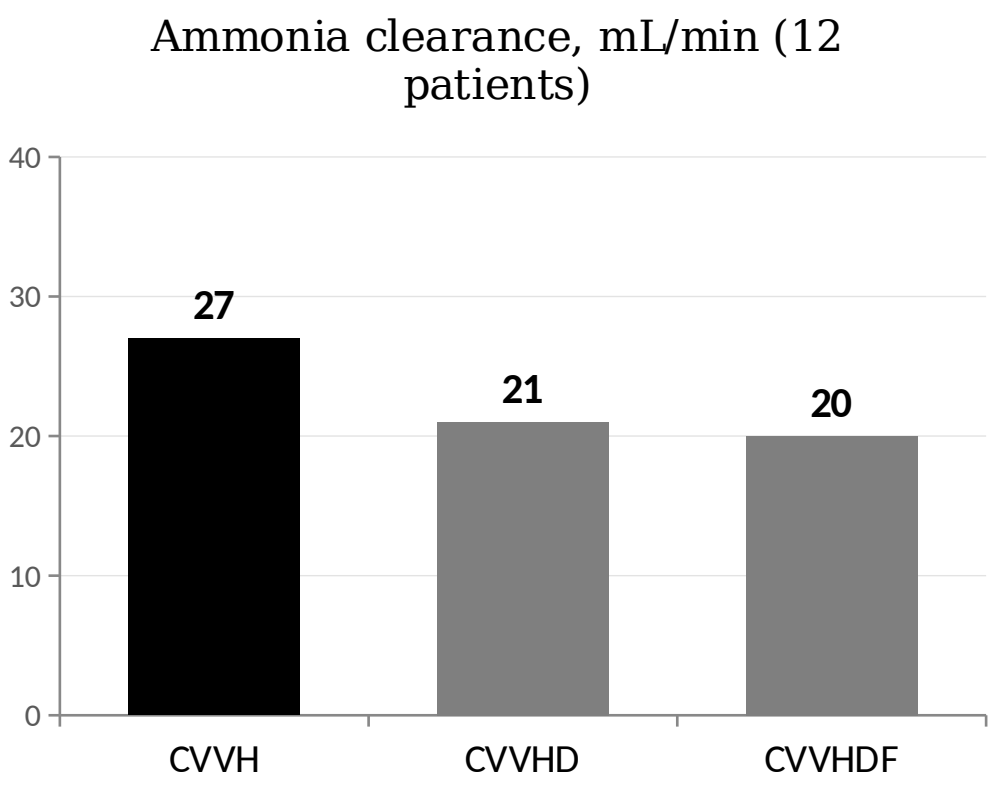
*I know what you're thinking...*

"CVVH doesn't clear ammonia"

**True or false?**

# "CVVH doesn't clear ammonia"

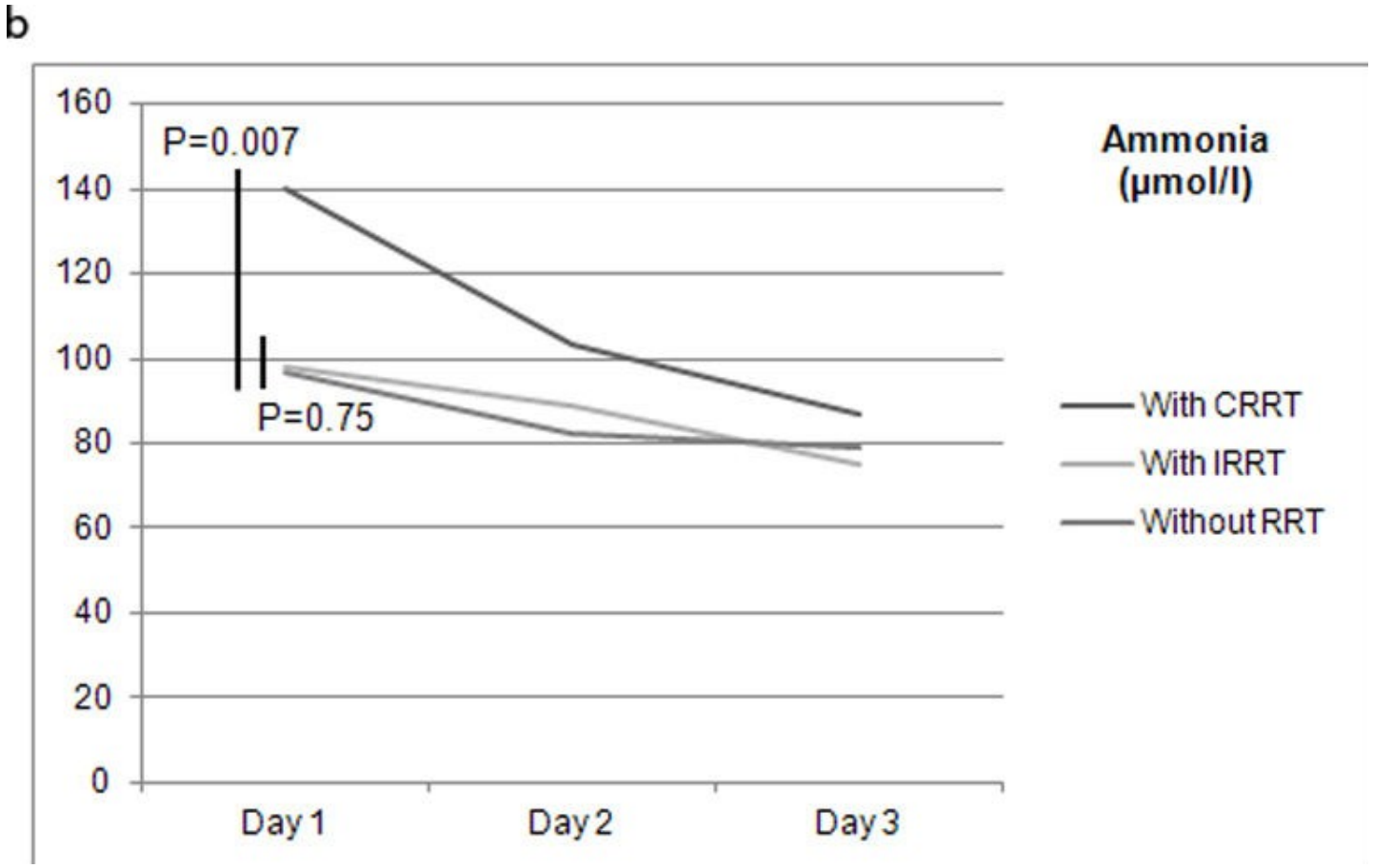
- Ammonia is tiny (17 Da), water soluble and not protein bound
- It crosses the filter with the water (convection, CVVH) or down its gradient (diffusion, CVVHD)
- So what matters is effluent flow and hours on a working circuit, not the letters
- More effluent, more clearance: **90 vs 35 mL/kg/h** (Slack 2014)
- 54 ALF patients on CRRT: median ammonia **151 → 52 µmol/L** by day 5 (Warrillow 2020)




**It does. About half the clearance of urea, so dose and filter life matter**

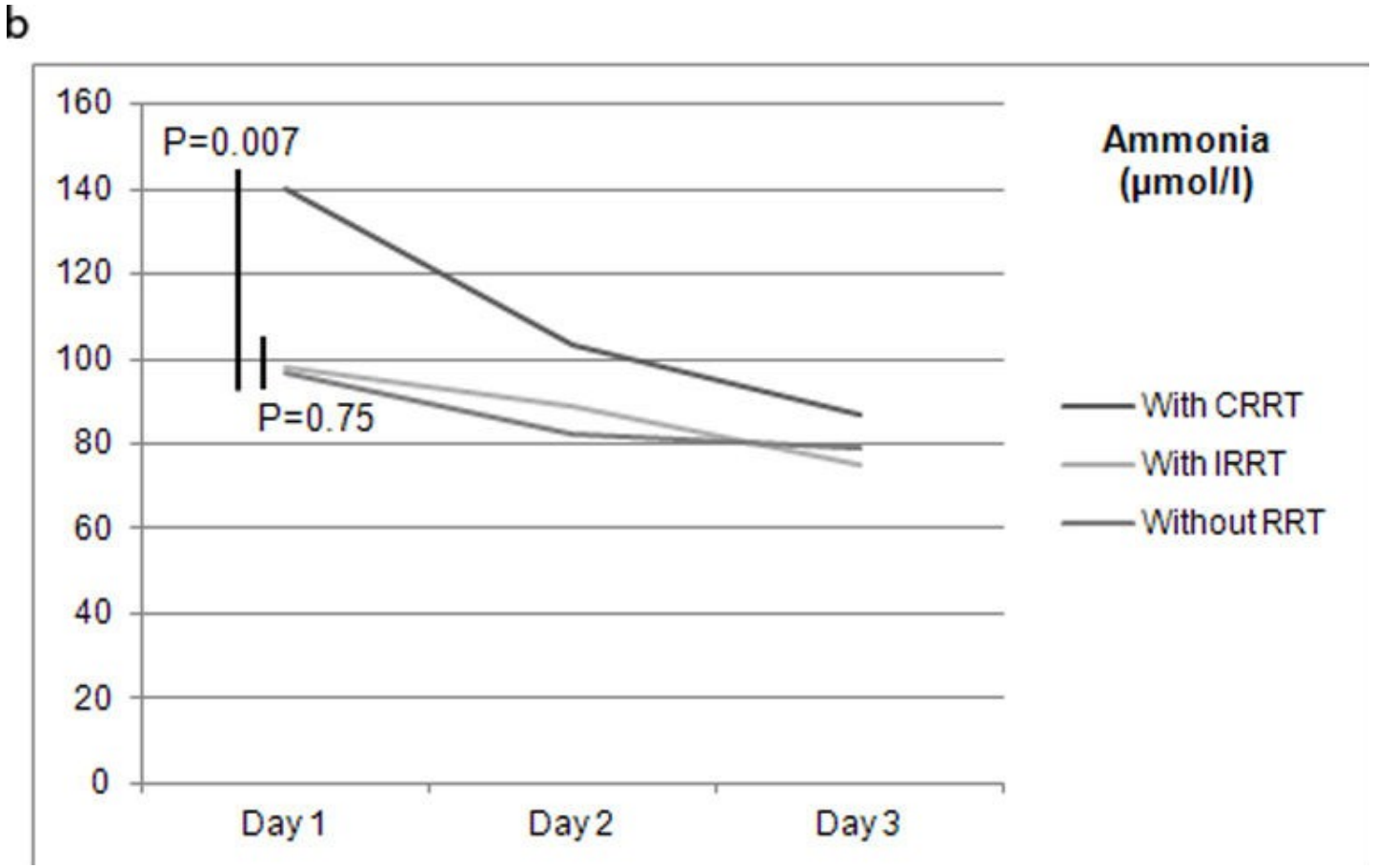
*Fisher C et al. Blood Purif 2022 [PMID: 35042216]; Warrillow SJ et al. Crit Care Med 2020 [PMID: 31939790]; Slack AJ et al. Liver Int 2014*

# Does early CRRT actually help?




*Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]*

# Does early CRRT actually help?




*Ammonia fell from about 140 to under 90 µmol/L over 3 days on CRRT. Little change with intermittent RRT or no RRT*

*Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]*

# Does early CRRT actually help?

| | **US ALF Study Group cohort (2018)** | **Pilot RCT (2026)** |
|---|---|---|
| Design | Large registry, observational | Open label, ALF with cerebral edema; both arms got plasma exchange |
| Comparison | CRRT vs no CRRT | CRRT within 12 hours vs CRRT only if plasma exchange failed |
| Finding | | |
| Catch | | |

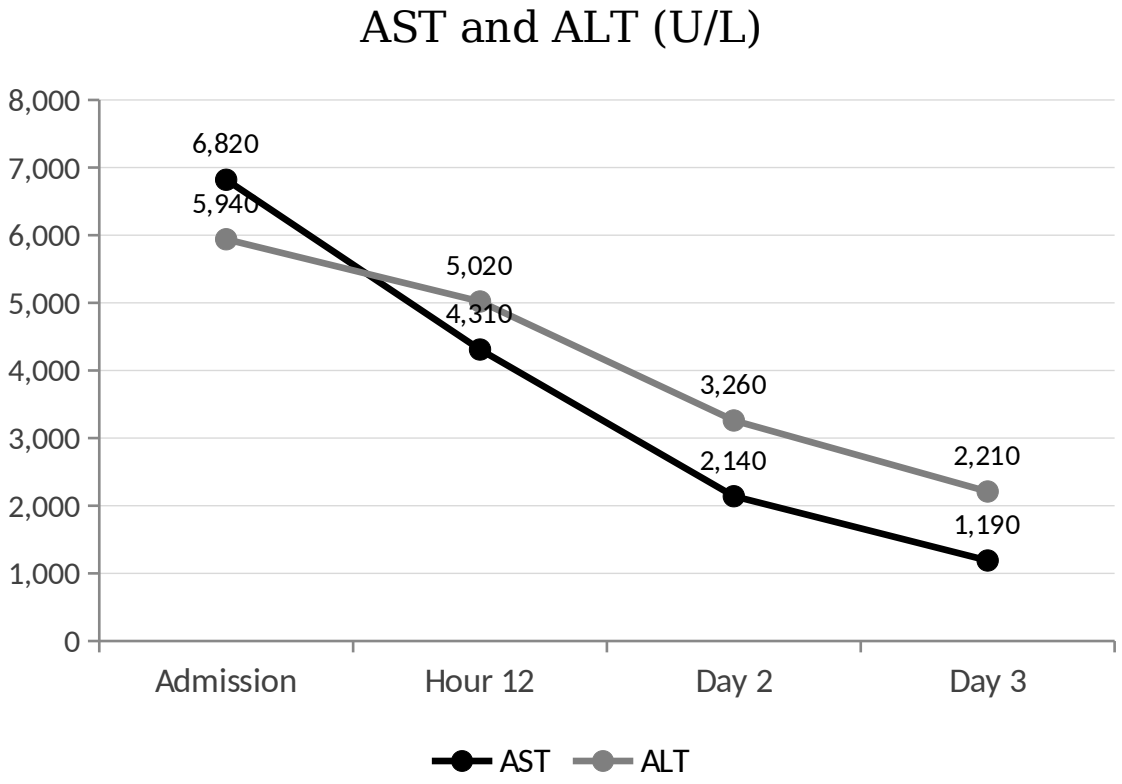
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

# Does early CRRT actually help?

| | **US ALF Study Group cohort (2018)** | **Pilot RCT (2026)** |
|---|---|---|
| Design | Large registry, observational | Open label, ALF with cerebral edema; both arms got plasma exchange |
| Comparison | CRRT vs no CRRT | CRRT within 12 hours vs CRRT only if plasma exchange failed |
| Finding | CRRT linked to lower ammonia and better 21-day transplant-free survival | Primary endpoint, 28-day survival: similar by intention to treat. Fewer deaths in the first 7 days (2.2% vs 17.8%) |
| Catch | **Association. Sicker or healthier patients may have been picked for CRRT** | **Small pilot, one center, many protocol violations. Needs a larger trial** |

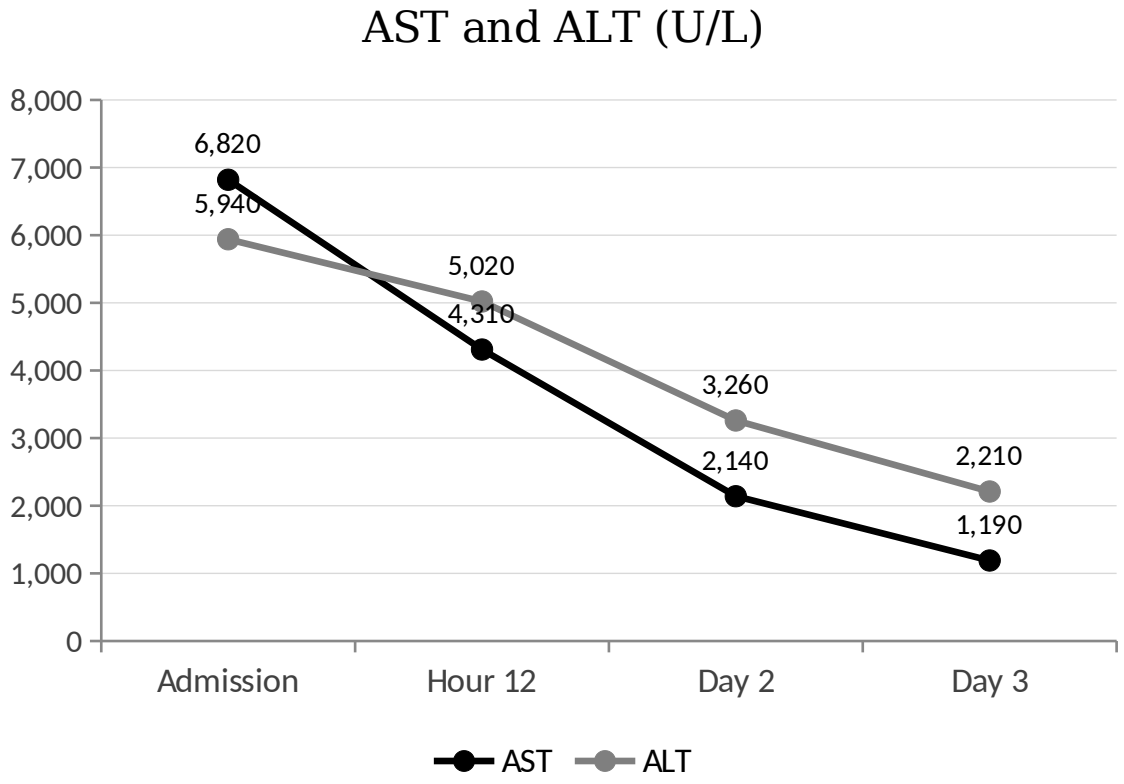
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

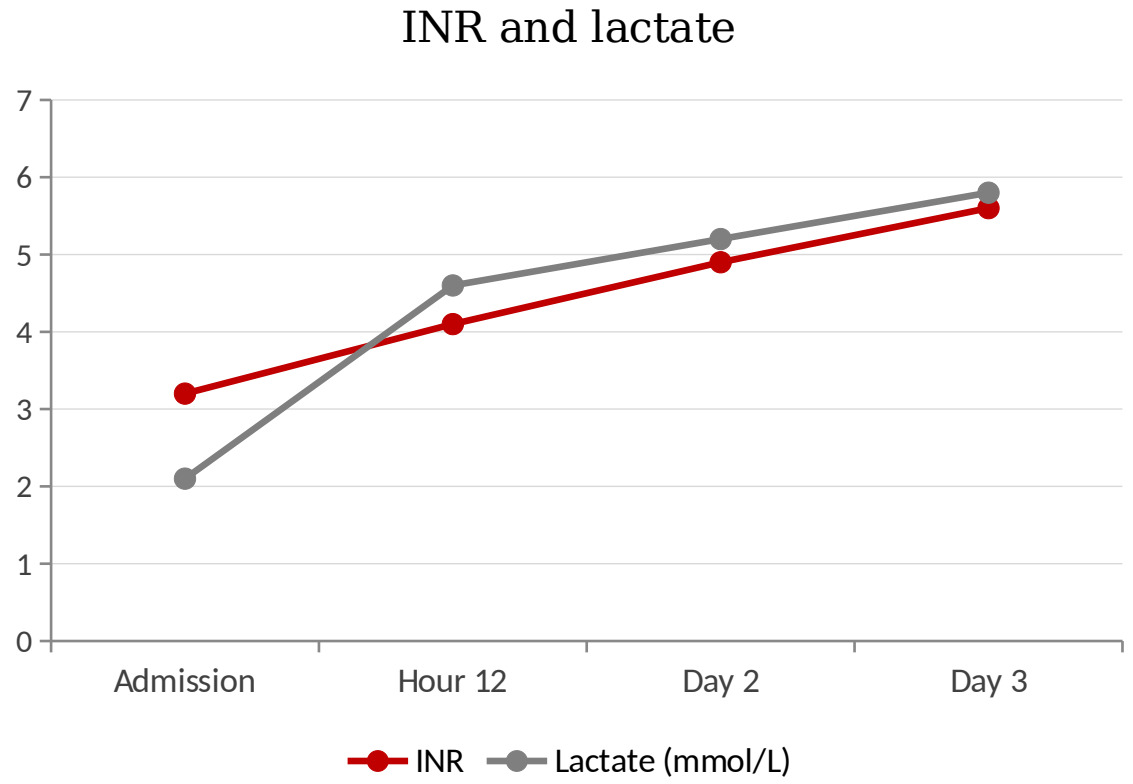
# Day 3: the enzymes are falling




**Is she getting better?**

# Day 3: the enzymes are falling




*Day 2 arterial pH 7.26 after fluids*

**No. The INR, the lactate and her brain are all worse**

*Made-up numbers for teaching*

# When do you call transplant hepatology?

**When she meets King's College criteria?**

**When MELD is high enough?**



# When do you call transplant hepatology?

**You should have called on admission, when you first suspected ALF**

- King's College criteria predict death without transplant. They are specific, not sensitive
- They are not a referral threshold. If you wait for them, the evaluation starts too late
- Patient A: arterial pH **7.26** on day 2 after fluids. She now meets the acetaminophen criteria



*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882]; full criteria in the appendix*

# Can plasma exchange buy time?

Research Article

JOURNAL OF HEPATOLOGY

**High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial**

Fin Stolze Larsen[1,*], Lars Ebbe Schmidt[1], Christine Bernsmeier[2], Allan Rasmussen[3], Helena Isoniemi[4], Vishal C. Patel[2], Evangelos Triantafyllou[2], William Bernal[2], Georg Auzinger[2], Debbie Shawcross[2], Martin Eefsen[1], Peter Nissen Bjerring[1], Jens Otto Clemmesen[1], Krister Hockerstedt[4], Hans-Jørgen Frederiksen[5], Bent Adel Hansen[1], Charalambos G. Antoniades[2,6,†], Julia Wendon[2,†]

## What do you think they found?

*Larsen FS et al. J Hepatol 2016;64:69-78 [PMID: 26325537]*

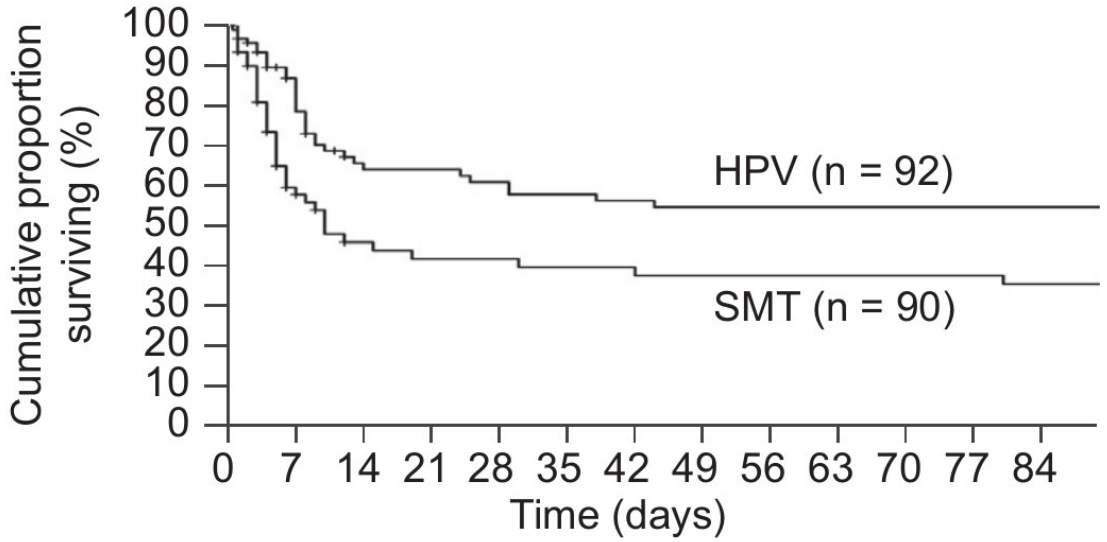
# Can plasma exchange buy time?

Research Article

JOURNAL OF HEPATOLOGY

## High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial

Fin Stolze Larsen[1,*], Lars Ebbe Schmidt[1], Christine Bernsmeier[2], Allan Rasmussen[3], Helena Isoniemi[4], Vishal C. Patel[2], Evangelos Triantafyllou[2], William Bernal[2], Georg Auzinger[2], Debbie Shawcross[2], Martin Eefsen[1], Peter Nissen Bjerring[1], Jens Otto Clemmesen[1], Krister Hockerstedt[4], Hans-Jørgen Frederiksen[5], Bent Adel Hansen[1], Charalambos G. Antoniades[2,6,†], Julia Wendon[2,†]




*Larsen FS et al. J Hepatol 2016;64:69-78 [PMID: 26325537]*

# Can plasma exchange buy time?

| | |
|---|---|
| Study | Open randomized trial, ALF, high-volume plasma exchange for 3 days vs standard care, 182 patients |
| Main finding | Hospital survival 58.7% vs 47.8% (HR 0.56); best in patients who were not transplanted |
| What it doesn't show | |
| So | |

*Larsen FS et al. J Hepatol 2016 [PMID: 26325537]*

# Can plasma exchange buy time?

| | |
|---|---|
| Study | Open randomized trial, ALF, high-volume plasma exchange for 3 days vs standard care, 182 patients |
| Main finding | Hospital survival 58.7% vs 47.8% (HR 0.56); best in patients who were not transplanted |
| What it doesn't show | **Not blinded; one regimen in one population. Says nothing about ACLF or about albumin dialysis devices** |
| So | A bridge to discuss with transplant hepatology. Not a reason to delay listing |

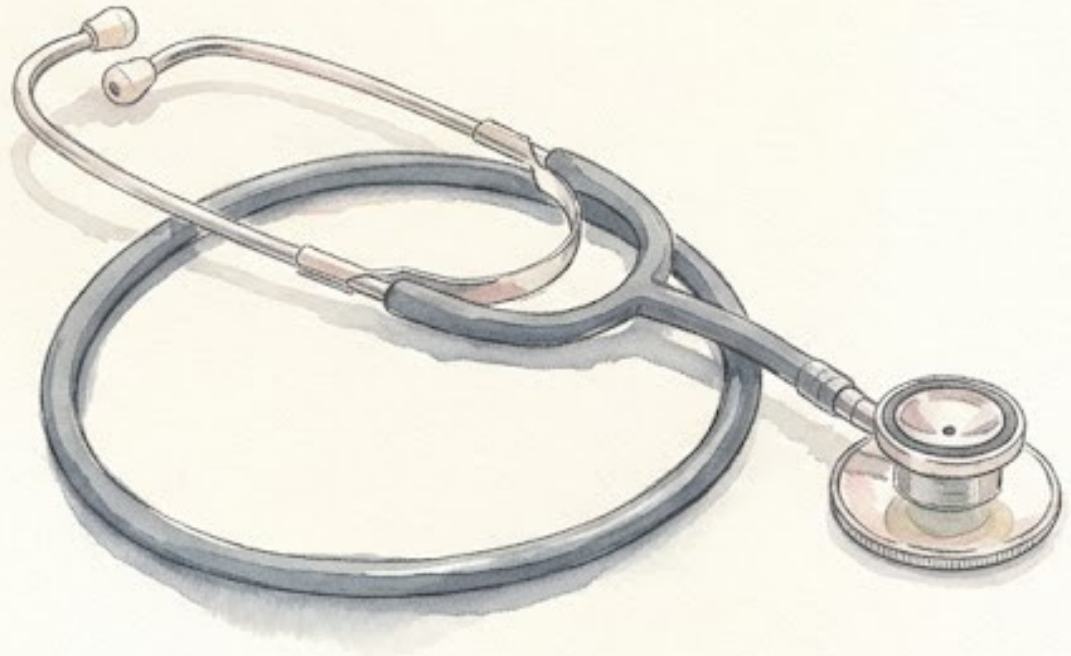
*Larsen FS et al. J Hepatol 2016 [PMID: 26325537]*

# Let's recap: bed 4, day 3

| Problem | What she's on | What we watch |
|---|---|---|
| Acetaminophen ALF, meets King's (pH 7.26) | NAC. Listed status 1A on day 2 | Time to an offer |
| Grade III-IV encephalopathy, ammonia 182 | Intubated, normal CO2, sodium 145 to 155, CRRT | Pupils, ammonia on CRRT, sodium |
| Lactate 5.8 and rising | Volume and MAP, look for infection | Lactate, MAP, pressor need |
| INR 5.6, not bleeding | No plasma | Bleeding, INR trend |
| Infection risk | Surveillance cultures | New fever or shock: low bar for antibiotics and antifungals |

*Made-up numbers for teaching*

# Any questions so far?



# Bed 7

56M, cirrhosis and ACLF



# Back to day 1: bed 7

| Bed 4 | Bed 7 |
| --- | --- |
| 28-year-old woman | 56-year-old man |
| Acetaminophen ALF | Alcohol-related cirrhosis with ascites |
| Intubated, on CRRT, transplant hepatology involved | Fever, confusion, low blood pressure |

**Is this liver failure too?**

# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

| | Now | Baseline |
|---|---|---|
| Bilirubin (mg/dL) | 5.8 | 2.1 |
| INR | 2.1 | 1.6 |
| Creatinine (mg/dL) | 2.6 | 0.9 |
| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

| | Now | Baseline |
|---|---|---|
| Bilirubin (mg/dL) | 5.8 | 2.1 |
| INR | 2.1 | 1.6 |
| Creatinine (mg/dL) | 2.6 | 0.9 |
| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

**Find the source: cultures, antibiotics, a diagnostic tap**

What tipped him over?

**INR 2.1, platelets 62,000. Do you tap him?**

# What tipped him over?

- **Yes, now. No plasma, no platelets**
- Ascitic fluid: **PMN 1,240/mm³**. Spontaneous bacterial peritonitis

**Common triggers**

- Infection: SBP, pneumonia, urine, lines, C. difficile
- Alcohol-associated hepatitis
- GI bleeding
- Drugs: NSAIDs, sedatives, too much diuretic
- HBV flare, ischemia
- No trigger is found in some patients. Repeat the search if he gets worse



*EASL ACLF 2023 [PMID: 37364789]; SCCM 2023 [PMID: 37052436]*

# Why is he hypotensive?

| Mechanism | What you do |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |

*Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]*

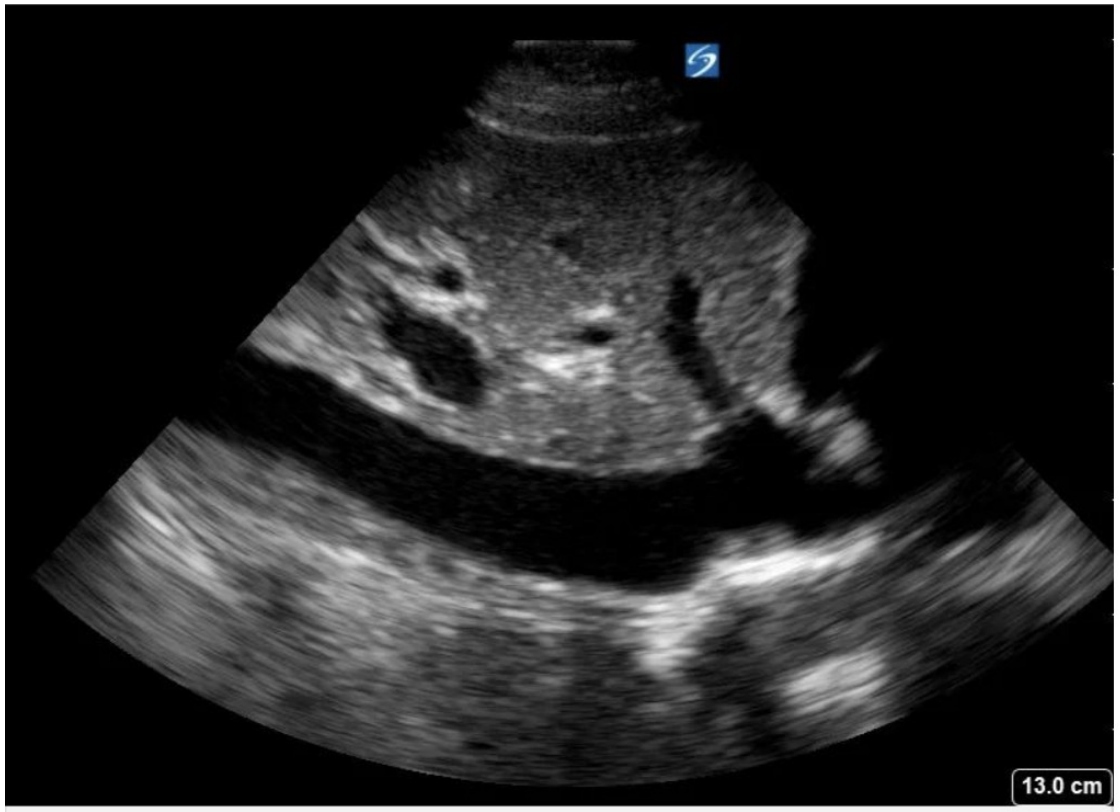
# Why is he hypotensive?

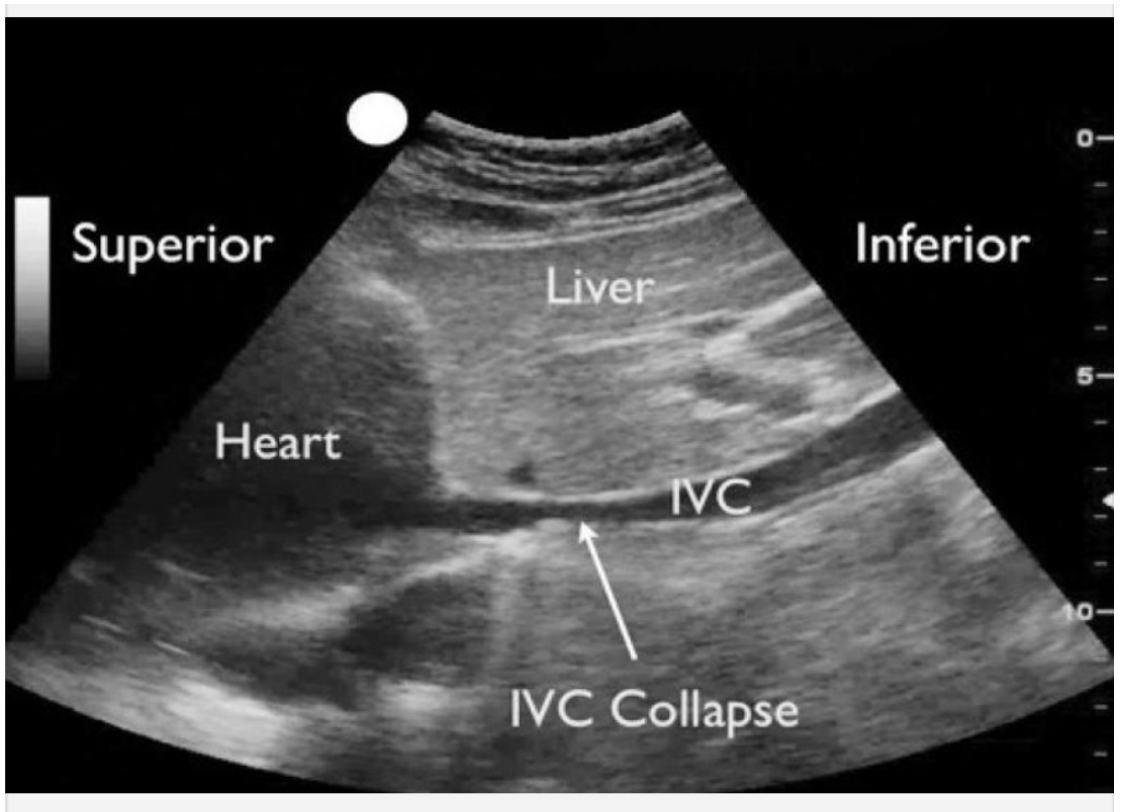
| Mechanism | What you do |
|---|---|
| Sepsis (SBP) | Cultures, antibiotics in the first hour, source control. Cover for prior exposure and local resistance |
| Splanchnic vasodilation of cirrhosis | Norepinephrine. Start at MAP 65 and adjust to perfusion |
| Too little volume, or too much | Ultrasound and dynamic measures. Edema does not mean the vessels are full |
| Hidden bleeding | Hemoglobin, rectal exam, think varices |
| Relative adrenal insufficiency | Hydrocortisone if shock is refractory |
| Cirrhotic cardiomyopathy | Echo |

**Usually more than one of these**

*Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]*

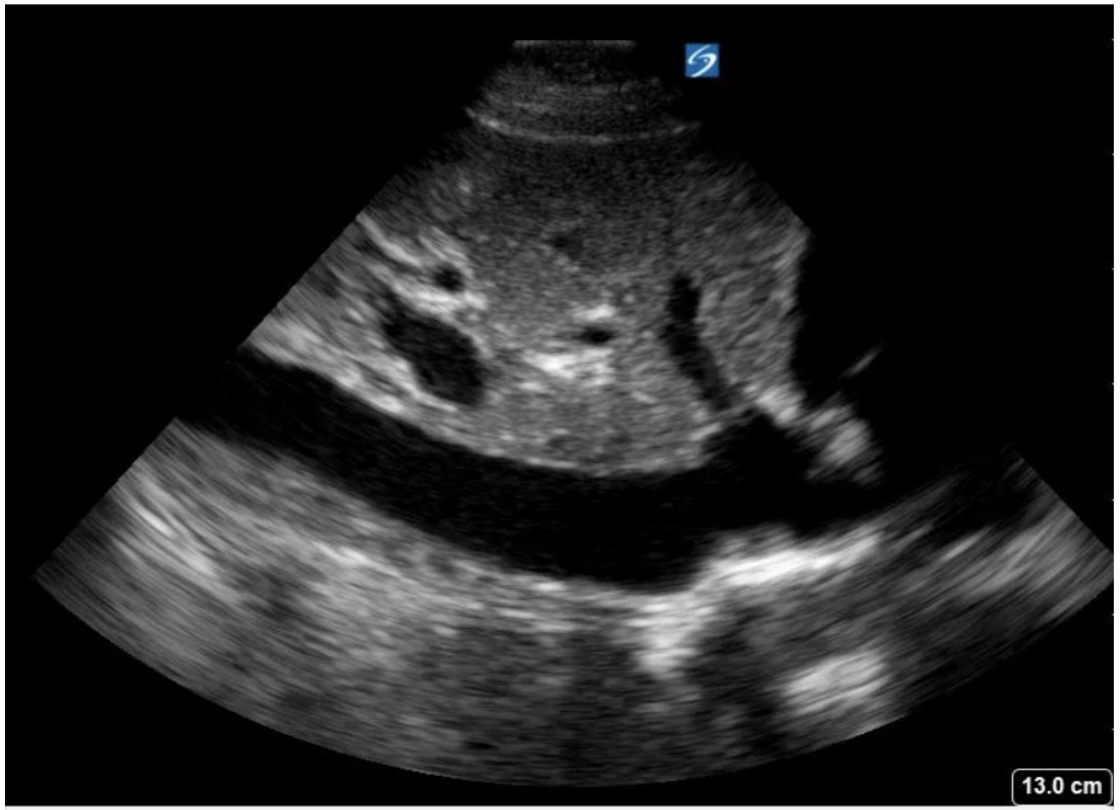
# Too little volume, or too much?

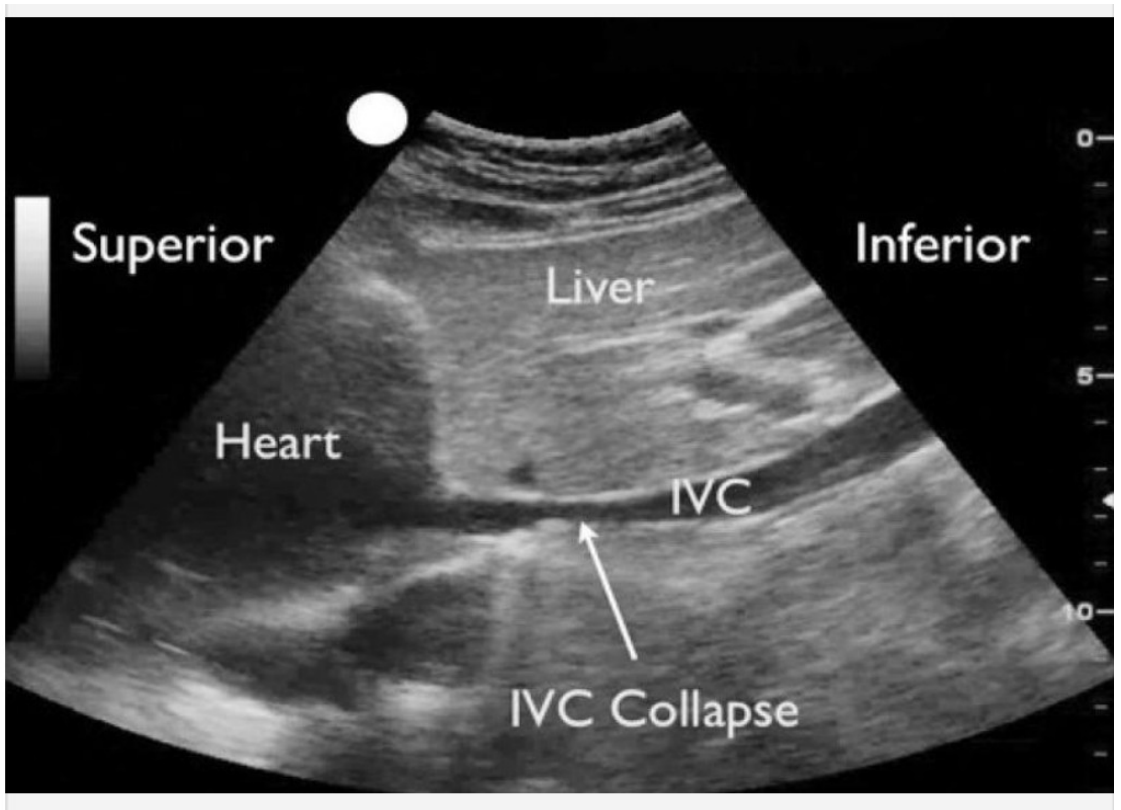



**Which one would you give more fluid?**

# Too little volume, or too much?

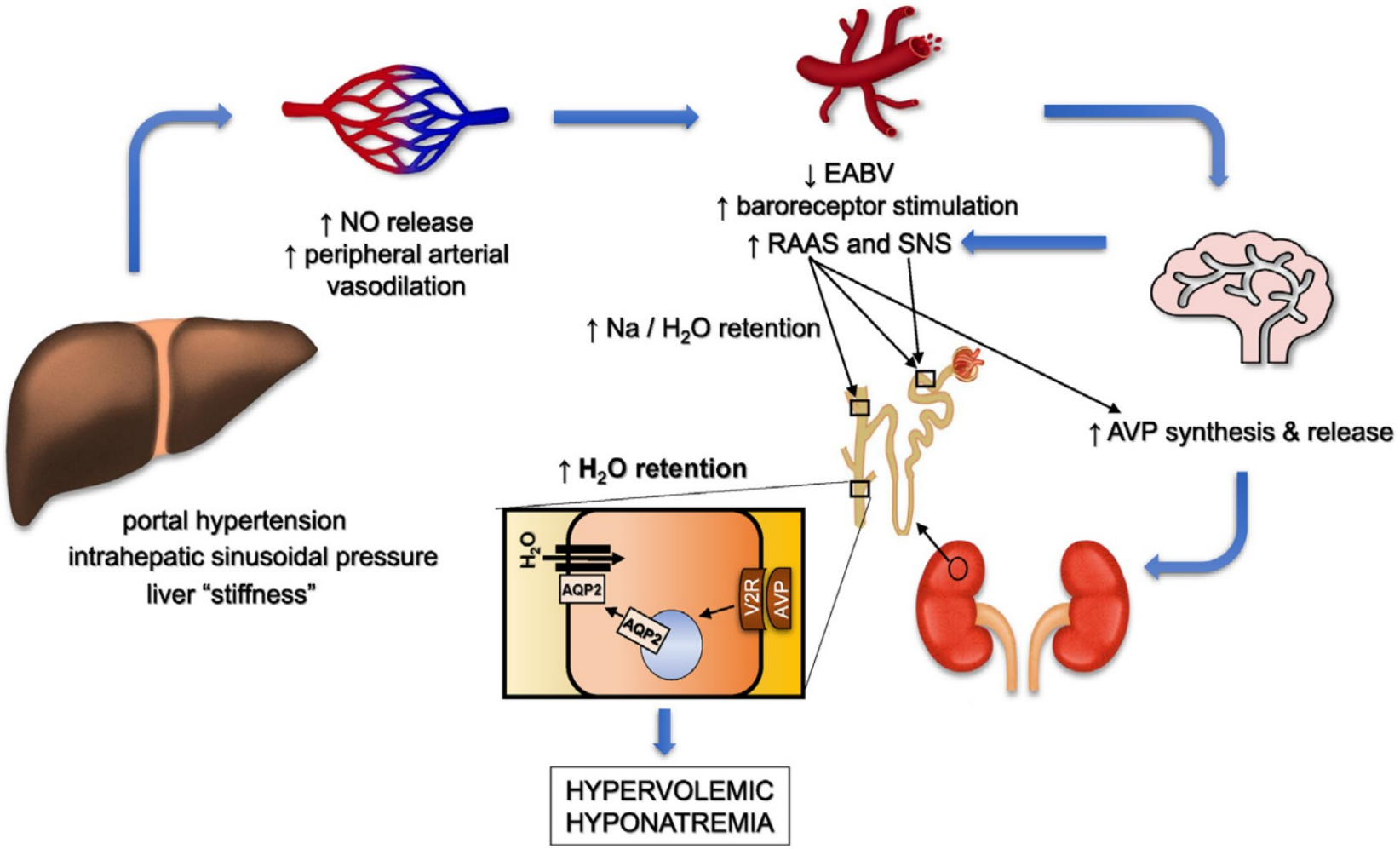



*Left: a full, plethoric IVC. Right: an IVC that collapses*

**Look before you give the next liter**

# Why is his pressure low to begin with?




*Rondon-Berrios H, Velez JCQ. Clin Liver Dis 2022*

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | | | |
| **What defines it** | | | |
| **What usually kills** | | | |
| **ICU focus** | | | |
| **Transplant** | | | |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | No cirrhosis | Cirrhosis | Cirrhosis (or chronic liver disease, depending on definition) |
| **What defines it** | INR ≥ 1.5 + encephalopathy | Ascites, bleeding, encephalopathy, jaundice | Decompensation + organ failure |
| **What usually kills** | Brain, multiorgan failure, sepsis | Bleeding, infection. Can turn into ACLF | Infection, shock, multiorgan failure |
| **ICU focus** | Brain, ammonia, early CRRT | Treat the trigger | Source control, shock, kidney vs lung |
| **Transplant** | Urgent listing pathway | Refer when MELD or the complication calls for it | Narrow window; reassess daily |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# Count the failing organs

| Organ | Failure (EASL-CLIF) |
|---|---|
| Liver | Bilirubin ≥ 12 mg/dL |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT |
| Brain | Encephalopathy grade III-IV |
| Coagulation | INR ≥ 2.5 |
| Circulation | Vasopressors (or terlipressin) |
| Lungs | PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy) |

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

# Count the failing organs

| Organ | Failure (EASL-CLIF) | Patient B |
|---|---|---|
| Liver | Bilirubin ≥ 12 mg/dL | 5.8: no |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT | **2.6: yes** |
| Brain | Encephalopathy grade III-IV | Grade II: no |
| Coagulation | INR ≥ 2.5 | 2.1: no |
| Circulation | Vasopressors (or terlipressin) | **Norepinephrine: yes** |
| Lungs | PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy) | PaO2/FiO2 211: no |

**Two organ failures: ACLF grade 2**

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

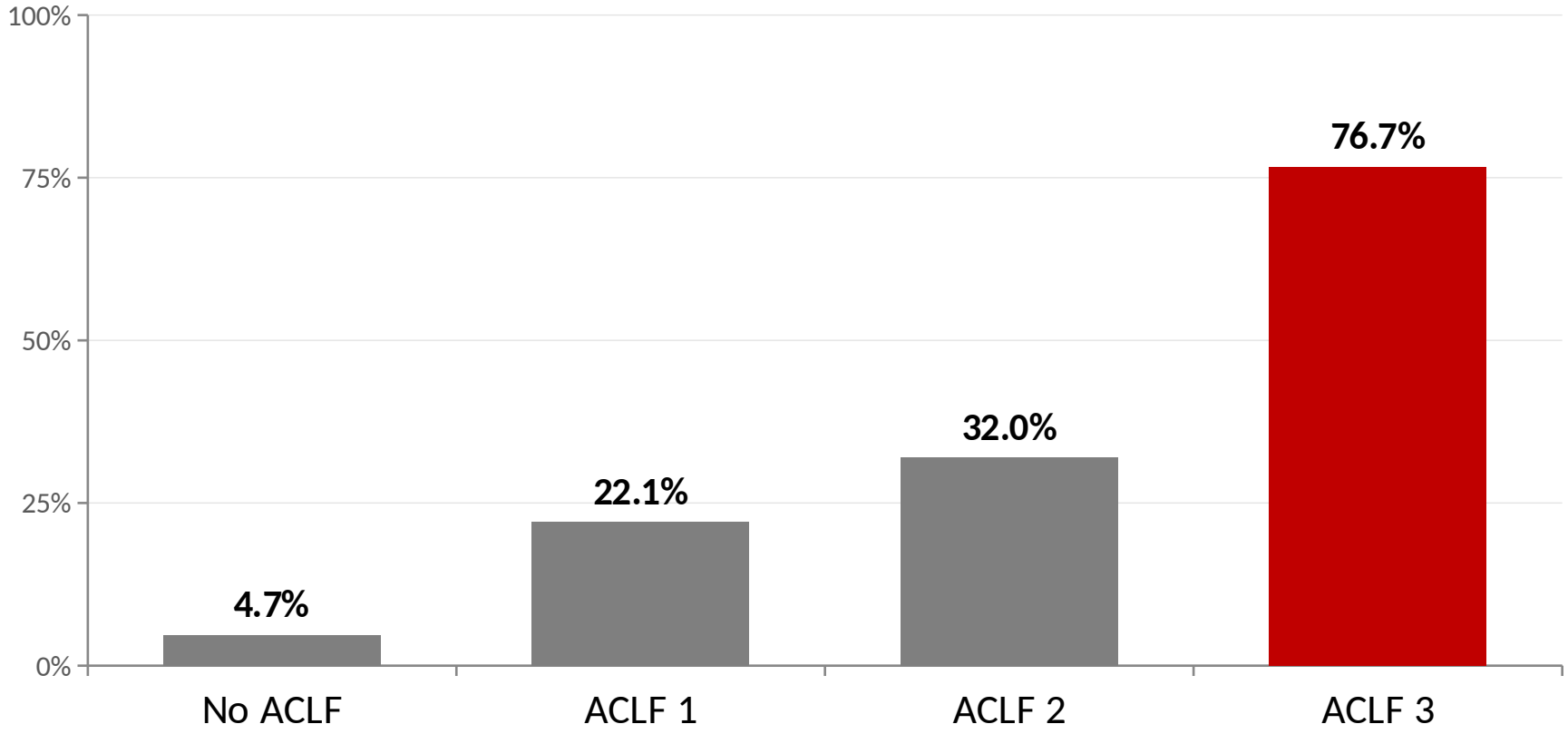
# ACLF grades

| Grade | What it takes |
|---|---|
| No ACLF | No organ failure; or<br>one failure of liver, clotting, circulation or lungs, with creatinine < 1.5 and no HE; or<br>brain failure alone, with creatinine < 1.5 |
| ACLF 1 | Kidney failure alone; or<br>one failure of liver, clotting, circulation or lungs, with creatinine 1.5 to 1.9 and/or grade I-II HE; or<br>brain failure alone, with creatinine 1.5 to 1.9 |
| ACLF 2 | Two organ failures |
| ACLF 3 | Three or more |

*A patient with "no ACLF" can still be very sick. Use the CLIF-C calculator*

*Moreau R et al. Gastroenterology 2013 [PMID: 23474284]; EASL ACLF 2023 [PMID: 37364789]*

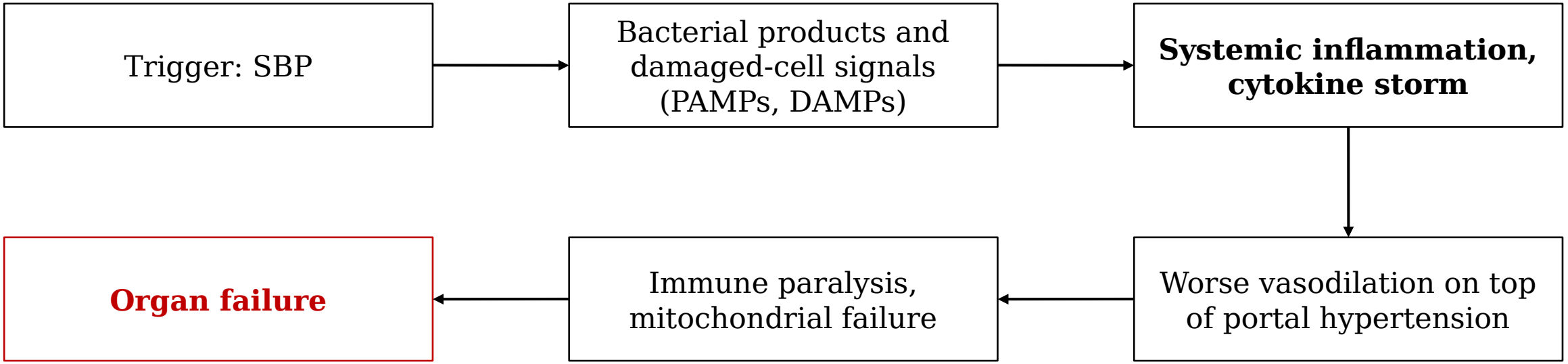
# 28-day mortality by ACLF grade




*CANONIC, 1,343 patients, 2013*

*Moreau R et al. Gastroenterology 2013 [PMID: 23474284]*

# Why do the organs fail?




- Bed 7 so far: **kidney** and **circulation**. Lungs and brain are next in line
- Treat the trigger early and support the organs; there is no drug for the inflammation itself

*Concept from Arroyo V, Moreau R, Jalan R. N Engl J Med 2020*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

**HRS? Start terlipressin?**

# HRS-AKI: what changed in 2024

| What many of us learned (ICA 2015) | Now (ADQI-ICA 2024) |
|---|---|
| No response after 2 days of diuretic withdrawal and albumin 1 g/kg<br>No shock<br>No nephrotoxins, no structural kidney disease | Cirrhosis with ascites<br>AKI by creatinine and/or urine output<br>No improvement within 24 hours of volume resuscitation, if he needs it<br>No albumin challenge if he is already euvolemic or wet<br>No strong alternative explanation as the main cause |

**Faster to call. But you still have to look for the other causes**

*Angeli P et al. J Hepatol 2015 (ICA); ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | | |
| AKI by creatinine and/or urine output | | |
| No improvement within 24 hours of volume resuscitation, if he needs it | | |
| No strong alternative explanation (shock, nephrotoxins, structural injury) | | |

**Go down the list for him. Which boxes does he tick?**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | Yes | ✅ |
| AKI by creatinine and/or urine output | Creatinine 0.9 → 3.0, urine 10 mL/h | ✅ |
| No improvement within 24 hours of volume resuscitation, if he needs it | 2 L given on arrival. Too early to say | ❌ |
| No strong alternative explanation (shock, nephrotoxins, structural injury) | **Septic shock on norepinephrine** | ❌ |

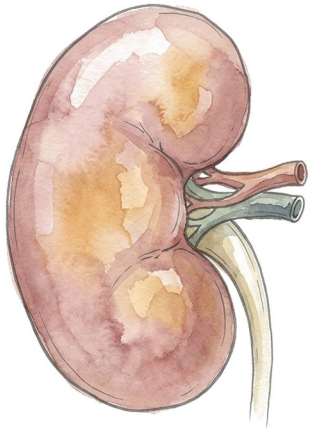
**Not HRS yet. Treat the shock first**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Kidney failure in cirrhosis is usually a mixture

**Possible contributors**

- Low effective volume
- Tubular injury: sepsis, contrast, drugs
- HRS physiology
- Tense ascites (abdominal compartment)
- Congestion

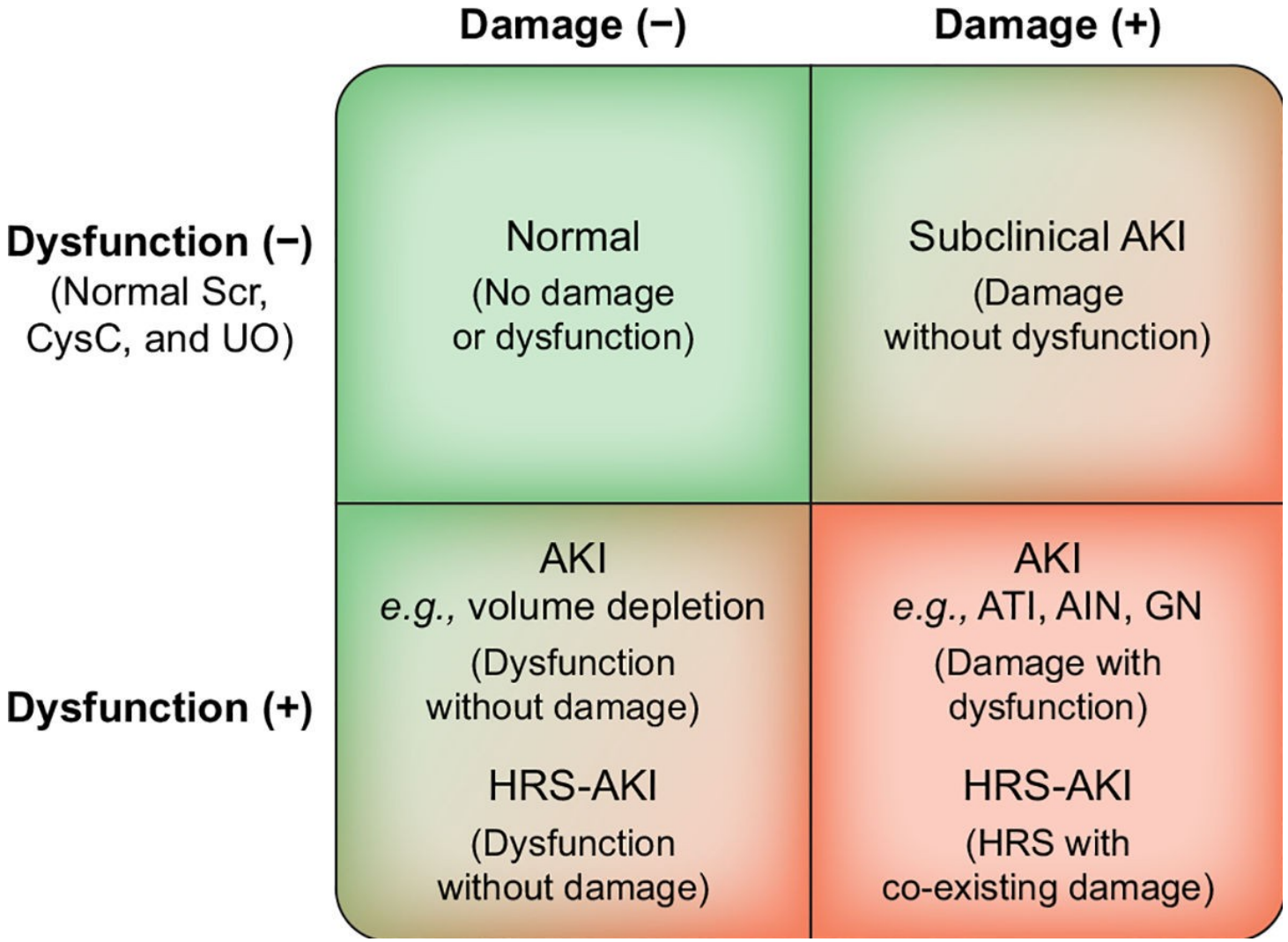


**Look**

- Urinalysis and sediment
- Drug list
- Ultrasound: kidneys, bladder, IVC, heart, lungs
- Bladder pressure if the belly is tight
- A low urine sodium is **not** proof of HRS

*ADQI-ICA consensus 2024 [PMID: 38527522]*

# Function and damage: four kinds of AKI




*Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 4 (CC BY 2.0, adapted from ADQI)*

# Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

**More lactulose? Repeat the ammonia in 6 hours?**

# Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

**Look for the reason first. The ammonia doesn't tell you**

- Glucose, sodium, oxygen and CO2
- Is the infection controlled? Cultures, source, antibiotic coverage
- Bleeding, sedatives and opioids, withdrawal, seizures, a new focal deficit (CT head)
- Lactulose for overt HE: 2 to 3 soft stools a day. By tube if he can't swallow safely; watch for dehydration and high sodium
- Rifaximin is an add-on, mainly to prevent the next episode
- Grade III-IV and a poor airway: intubate
- No repeat ammonia

*EASL HE guideline 2022 [PMID: 35724930]; SCCM 2023 [PMID: 37052436]*

# Day 2: worse kidney, worse lungs

| | Admission | Day 2 |
|---|---|---|
| Albumin received | | 110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid |
| Breathing | SpO2 92%, Venturi 35% | **SpO2 86% on Venturi 50%, RR 32, crackles** |
| Ultrasound | | Diffuse B-lines, IVC full |
| Creatinine (mg/dL) | 2.6 | 3.1 |
| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

**More albumin? Terlipressin? Something else?**

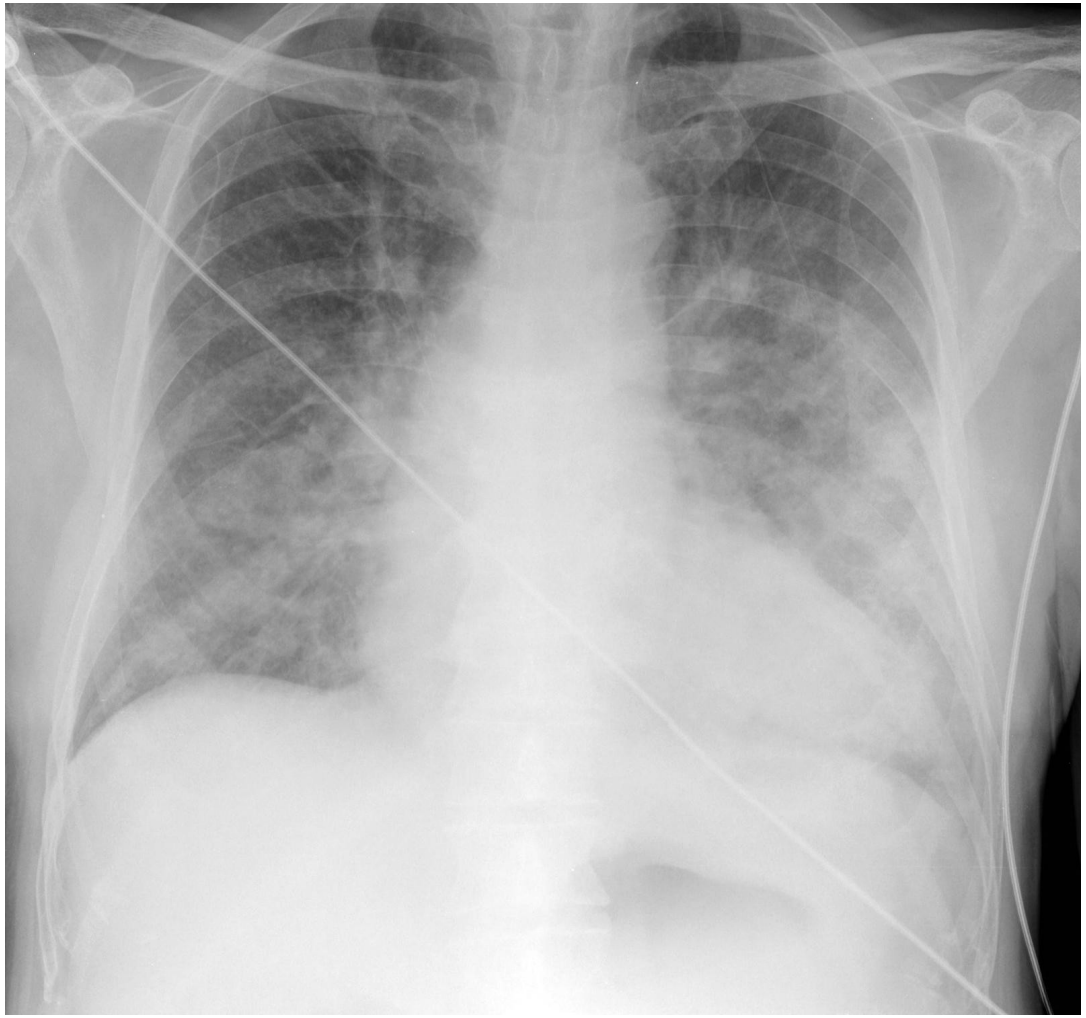
# Day 2: worse kidney, worse lungs

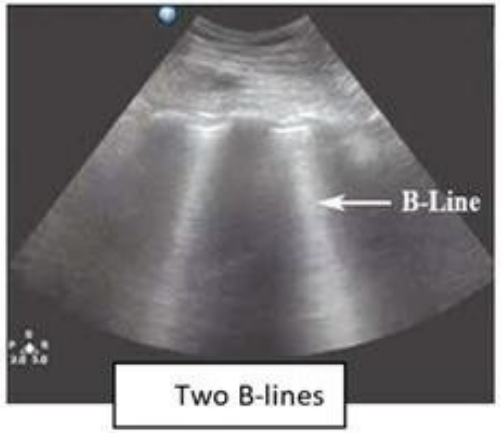
| | Admission | Day 2 |
|---|---|---|
| Albumin received | | 110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid |
| Breathing | SpO2 92%, Venturi 35% | **SpO2 86% on Venturi 50%, RR 32, crackles** |
| Ultrasound | | Diffuse B-lines, IVC full |
| Creatinine (mg/dL) | 2.6 | 3.1 |
| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

**First, what are the lungs telling you?**

**Now on high-flow, FiO2 0.6, SpO2 90%: SpO2/FiO2 150. That's a third organ failure**

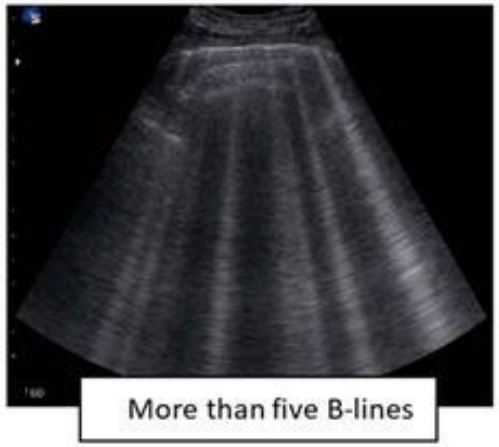
# Day 2: what do his lungs look like?

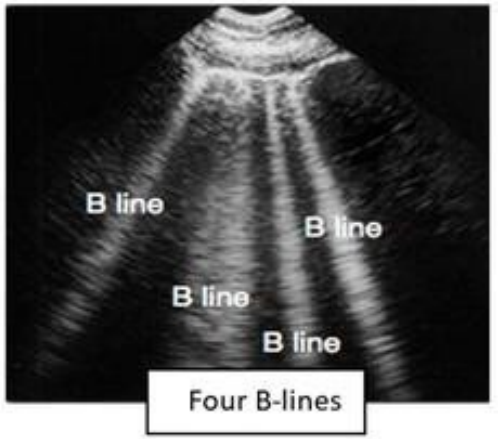











*Images for teaching, not this patient*

# Will fixing the kidney hurt the lungs?

**Terlipressin, more albumin, dialysis: what would you do for him today?**

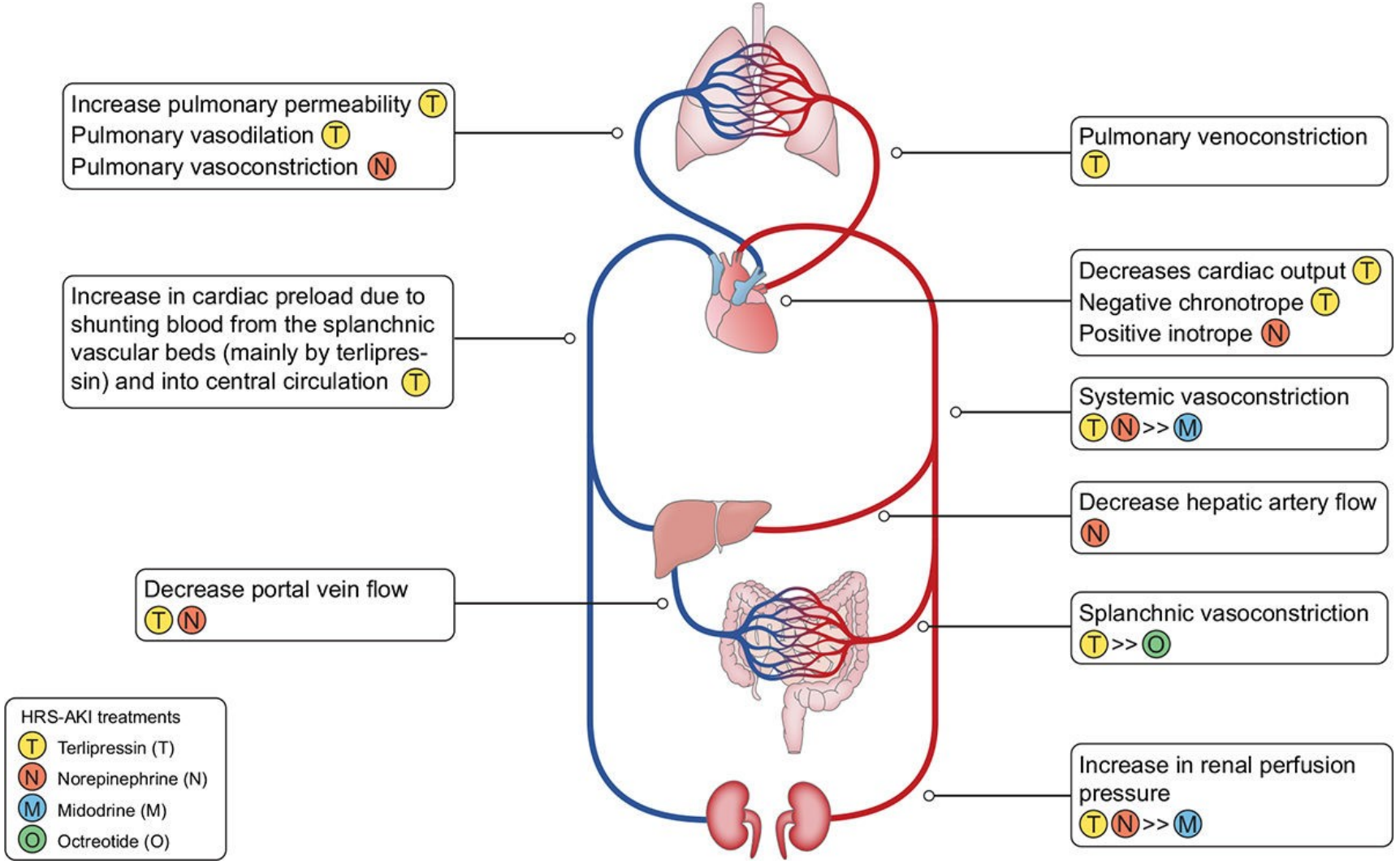
# Will fixing the kidney hurt the lungs?

- **Terlipressin: not for him today**
  - Label: contraindicated with hypoxia or worsening respiratory symptoms; don't start if SpO2 < 90%, stop if it falls below 90%
  - Label: avoid in ACLF grade 3; volume overload raises the risk of respiratory failure
- Stop the albumin while he is wet
- He is already on a vasoconstrictor. Norepinephrine, titrated to perfusion
- RRT if he stays overloaded. For HRS, RRT is a bridge to transplant: if he isn't a candidate, talk about goals before you start
- Midodrine and octreotide? They work poorly, and not in someone on IV pressors

**Check oxygen and volume before any kidney treatment**

*TERLIVAZ prescribing information, revised 6/2026; ADQI-ICA 2024 [PMID: 38527522]*

# What each HRS drug does to the heart and lungs




*Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 5 (CC BY 2.0, adapted from ADQI)*

# Does HRS reversal mean survival?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

**What do you think they found?**

*Wong F et al. N Engl J Med 2021;384:818-28 [PMID: 33657294]*

# Does HRS reversal mean survival?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

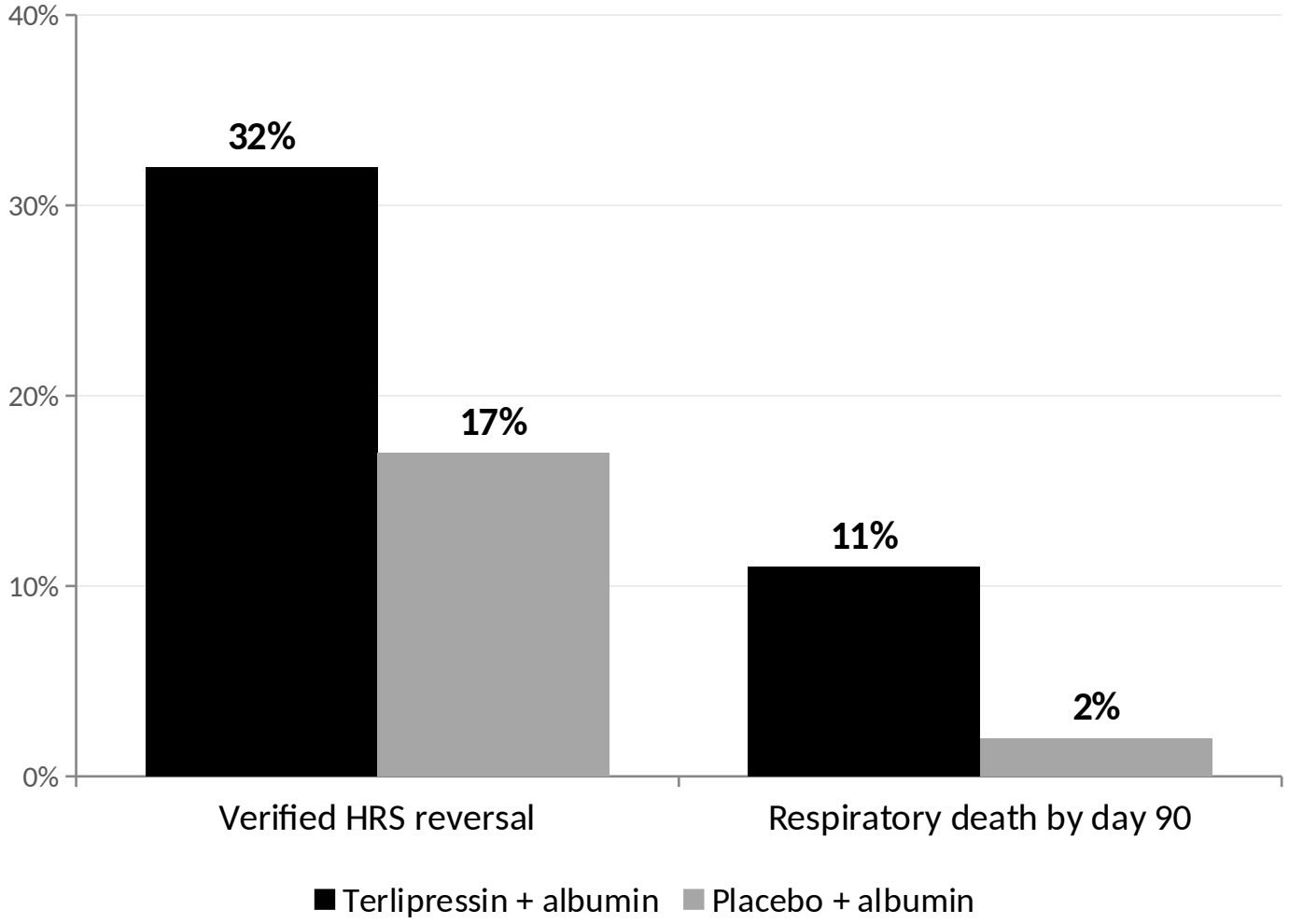
F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

**CONCLUSIONS**

In this trial involving adults with cirrhosis and HRS-1, terlipressin was more effective than placebo in improving renal function but was associated with serious adverse events, including respiratory failure. (Funded by Mallinckrodt Pharmaceuticals; CONFIRM ClinicalTrials.gov number, NCT02770716.)

*Wong F et al. N Engl J Med 2021;384:818-28 [PMID: 33657294]*

# Does HRS reversal mean survival?




**CONFIRM: type 1 HRS, 300 patients**

- More kidneys recovered
- **More patients died of respiratory failure**
- All-cause 90-day mortality was not different
- Respiratory failure as an adverse event: about 14% vs 5% (label)

*Wong F et al. NEJM 2021 [PMID: 33657294]; TERLIVAZ prescribing information, revised 6/2026*

# Should we treat the albumin number?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## A Randomized Trial of Albumin Infusions in Hospitalized Patients with Cirrhosis

Louise China, Ph.D., Nick Freemantle, Ph.D., Ewan Forrest, M.D., Yiannis Kallis, Ph.D., Stephen D. Ryder, D.M., Gavin Wright, Ph.D., Andrew J. Portal, M.D., Natalia Becares Salles, Ph.D., Derek W. Gilroy, Ph.D., and Alastair O'Brien, Ph.D., for the ATTIRE Trial Investigators*

**What do you think they found?**

*China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]*

# Should we treat the albumin number?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## A Randomized Trial of Albumin Infusions in Hospitalized Patients with Cirrhosis

Louise China, Ph.D., Nick Freemantle, Ph.D., Ewan Forrest, M.D., Yiannis Kallis, Ph.D., Stephen D. Ryder, D.M., Gavin Wright, Ph.D., Andrew J. Portal, M.D., Natalia Becares Salles, Ph.D., Derek W. Gilroy, Ph.D., and Alastair O'Brien, Ph.D., for the ATTIRE Trial Investigators*

**CONCLUSIONS**

In patients hospitalized with decompensated cirrhosis, albumin infusions to increase the albumin level to a target of 30 g per liter or more was not more beneficial than the current standard care in the United Kingdom.

*China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]*

# Should we treat the albumin number?

| | |
|---|---|
| Study | ATTIRE: daily albumin to keep serum albumin ≥ 30 g/L vs standard care, hospitalized decompensated cirrhosis, 777 patients |
| Main finding | Composite of new infection, kidney dysfunction or death: 29.7% vs 30.2% |
| The catch | **More serious adverse events with albumin, including fluid overload and pulmonary edema** |

**When albumin is indicated**

- SBP: 1.5 g/kg on day 1, 1 g/kg on day 3
- Paracentesis over 5 L: 6 to 8 g per liter removed
- HRS-AKI: with a vasoconstrictor, dosed to volume status

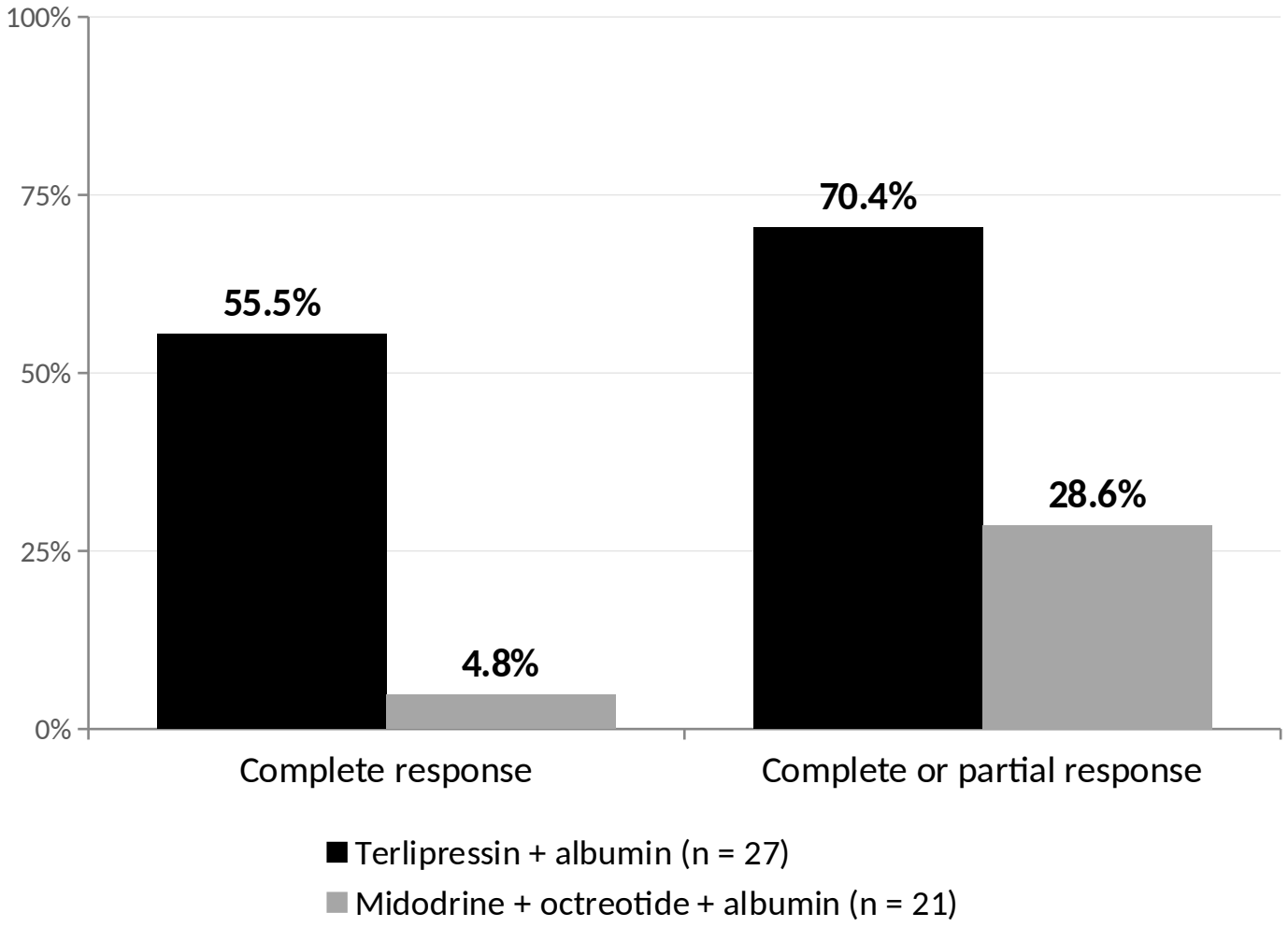
*China L et al. NEJM 2021 [PMID: 33657293]; ADQI-ICA 2024 [PMID: 38527522]*

# Midodrine and octreotide for HRS-AKI?

Head to head: terlipressin vs midodrine plus octreotide, both with albumin, type 1 HRS

**How different can two vasoconstrictors be?**

# Midodrine and octreotide for HRS-AKI?




**Randomized, 8 Italian centers, type 1 HRS. 49 randomized (27 vs 22), 48 analyzed**

- Stopped early at the interim analysis because the difference was already clear
- Blood pressure rose more with terlipressin
- 3-month survival 59% vs 43%; too small to tell
- **Use it only if terlipressin and norepinephrine are not options**

*Cavallin M et al. Hepatology 2015;62:567-74 [PMID: 25644760]; ADQI-ICA 2024 [PMID: 38527522]*

# What changed since yesterday?

| | Day 1 | Day 2 | Day 3 | Day 5 |
|---|---|---|---|---|
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 | 0.12 | Off |
| Oxygen, SpO2 | Venturi 0.35, 92% | HFNC 0.6, 90% | HFNC 0.4, 95% | 2 L, 97% |
| Lung failure? | No (P/F 211) | Yes (S/F 150) | No (S/F 238) | No |
| Kidney | Cr 2.6 | Cr 3.1 | CRRT for fluid | On CRRT |
| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | ? |

**Better or worse? What do you tell his family?**

*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

# What changed since yesterday?

| | Day 1 | Day 2 | Day 3 | Day 5 |
|---|---|---|---|---|
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 | 0.12 | Off |
| Oxygen, SpO2 | Venturi 0.35, 92% | HFNC 0.6, 90% | HFNC 0.4, 95% | 2 L, 97% |
| Lung failure? | No (P/F 211) | Yes (S/F 150) | No (S/F 238) | No |
| Kidney | Cr 2.6 | Cr 3.1 | CRRT for fluid | On CRRT |
| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | **1** |

- The grade at days 3 to 7 predicts outcome better than the grade at admission
- Persistent ACLF 3 at days 3 to 7, especially with four or more organ failures, has a very high mortality without transplant
- If he is not a transplant candidate, discuss goals of care with the family and involve palliative care

*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

# Who gets a liver?

Transplant in ACLF



# Too sick for a transplant, or sick enough to need one?

**Ventilator, pressors, CRRT, ACLF 3. Is he too sick?**

# Too sick for a transplant, or sick enough to need one?

- A ventilator, pressors or dialysis do not by themselves exclude transplant
- Selected patients with ACLF grade 3 do well after transplant. The window is short

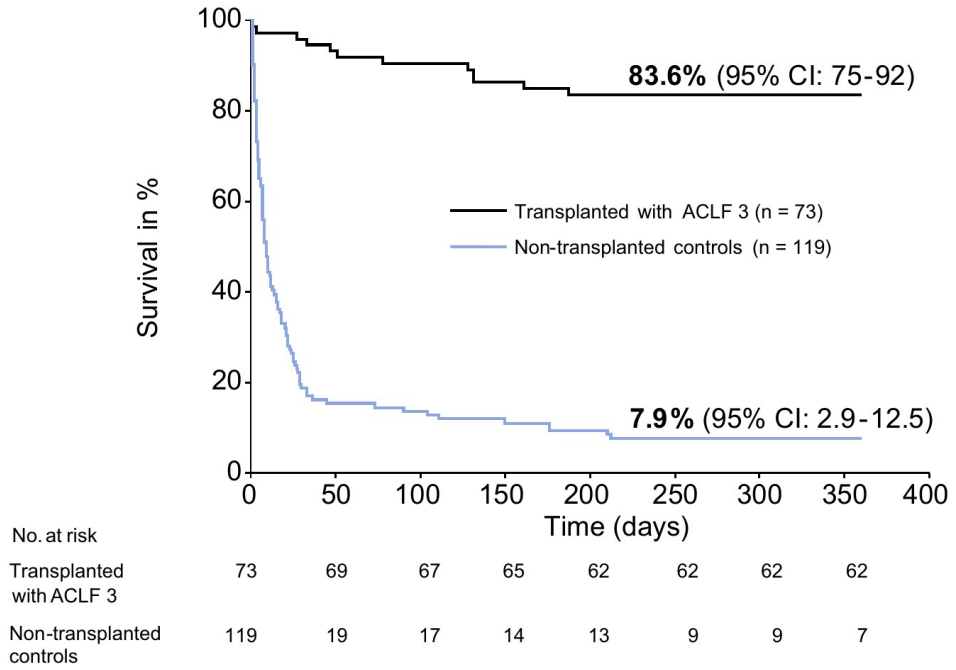
**What can close the window**

- Uncontrolled infection
- Rising vasopressor needs, severe respiratory failure
- Irreversible brain injury
- Frailty and comorbidity
- Alcohol: centers now evaluate individually rather than by a fixed abstinence rule

Research Article

EASL JOURNAL OF HEPATOLOGY

**Liver transplantation in the most severely ill cirrhotic patients: A multicenter study in acute-on-chronic liver failure grade 3**




*Selected patients, not randomized*

*Artru F et al. J Hepatol 2017 [PMID: 28645736]; Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]; AASLD/AST 2026 [PMID: 41405234]*

# Calling transplant hepatology

**Call first. Gather this while you wait for the call back**

| Patient A (ALF) | Patient B (ACLF) |
|---|---|
| Cause and time course (days from jaundice to encephalopathy)<br>Encephalopathy grade, pupils, sedation<br>Airway and ventilator settings<br>Ammonia, sodium, CRRT<br>INR, lactate, pH, glucose trends<br>Pressors, infection<br>Social support, intent of ingestion | Cause of cirrhosis, recent alcohol use<br>Precipitant and source control<br>Organ failures today and yesterday<br>Pressor dose, oxygen, RRT<br>MELD, CLIF-C scores<br>Frailty, comorbidities<br>Prior transplant evaluation, support |

# ALF and ACLF: what changes at the bedside

*Both have a sick liver. Do they have the same disease?*

| | Acute liver failure | ACLF |
|---|---|---|
| **Ammonia** | | |
| **Lactulose** | | |
| **Sodium** | | |
| **Antibiotics** | | |
| **INR** | | |

*ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]*

# ALF and ACLF: what changes at the bedside

*Both have a sick liver. Do they have the same disease?*

| | Acute liver failure | ACLF |
|---|---|---|
| **Ammonia** | With the HE grade and the trend: brain risk, and when to start CRRT. No level rules out ICH | Doesn't diagnose HE. Don't trend it to dose lactulose |
| **Lactulose** | No proven benefit; bowel distension can get in the way of surgery | Yes for overt HE, plus treat the trigger. Rifaximin as an add-on. Avoid dehydration |
| **Sodium** | Grade III-IV or high ammonia: center protocol, often 145 to 155, matched to the CRRT fluid | Often chronically low: correct slowly. In high-risk patients, no more than 8 mmol/L in 24 hours |
| **Antibiotics** | No routine prophylaxis. Surveillance cultures, low threshold to treat | Treat infection fast; prophylaxis with GI bleeding |
| **INR** | Tracks severity, poor at predicting bleeding. No plasma just for the number | Same. Judge the procedure and the patient |

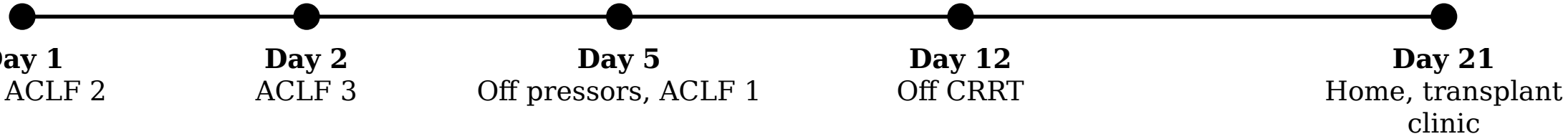
*ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]*

# What happened to them

## Bed 4




## Bed 7




*Made-up outcomes for teaching*

# Back to our patients

- Which one needed transplant hepatology before the tests came back?
- Why did a falling ALT make you more worried?
- When is CRRT right before the creatinine is high?
- Why isn't lactulose for ACLF evidence for ALF?
- Why were albumin and terlipressin the wrong next step for bed 7?
- What would make you change yesterday's plan?

# Some tricks of the trade

- High AST and ALT alone are not liver failure. Get an INR and examine the patient
- Stabilize, treat the cause and call transplant hepatology in the same hour
- ALF: **protect the brain**. Intubate at grade III, keep the CO2 normal, follow your center's sodium plan, and start CRRT for rising ammonia
- ACLF: count the organs, find the trigger
- ACLF: check oxygenation and volume status before albumin or terlipressin
- Reassess organ failures daily. The trend matters more than the admission score



# Questions?

# Appendix: West Haven grades

| Grade | What you see |
|---|---|
| I | Short attention, mild confusion, reversed sleep, subtle personality change |
| II | Lethargy, disorientation to time, obvious personality change, asterixis |
| III | Somnolent but rousable to voice or pain, grossly disoriented, bizarre behavior |
| IV | Coma |

# Appendix: King's College criteria

| Acetaminophen | Not acetaminophen |
|---|---|
| **Arterial pH < 7.30 after fluids**<br>*or all three within 24 hours:*<br>INR > 6.5<br>Creatinine > 3.4 mg/dL<br>Grade III-IV encephalopathy<br>*(Modified: lactate > 3.5 at 4 h or > 3.0 at 12 h after fluids)* | **INR > 6.5**<br>*or any three of:*<br>Age < 10 or > 40<br>Drug reaction, or indeterminate cause<br>Jaundice to encephalopathy > 7 days<br>INR > 3.5<br>Bilirubin > 17.5 mg/dL |

*Original O'Grady criteria; the lactate line is the later modification*

*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882] and update 2022 [PMID: 36001996]*

# Appendix: arterial ammonia studies in ALF

| Study (arterial ammonia) | Patients | What they found | The catch |
|---|---|---|---|
| Clemmesen 1999, Copenhagen | 44 ALF | **14 herniated. Mean ammonia 230 vs 118 µmol/L; every herniation was above 146** | Small, one center |
| Bhatia 2006, Delhi | 80 ALF | Admission ammonia ≥ 124 predicted death (sensitivity 79%, specificity 76%) | One center; the endpoint was death, not ICH |
| Bernal 2007, King's | 165 ALF, grade III-IV | > 100 predicted severe HE (70% accuracy). **> 200: ICH in 55%** | 200 missed most of the ICH cases |
| Kumar 2012, Delhi | 295 ALF | **≥ 122 for 3 days: survival 23% vs 72%**, cerebral edema 71% vs 37% | One center, but the trend beat a single value |

**Above 150 to 200 and not falling: worry about the brain**

*ICH can occur below 200. Interpret the level with the HE grade and the trend*

*Clemmesen 1999 [PMID: 10051463]; Bhatia 2006 [PMID: 16024550]; Bernal 2007 [PMID: 17685471]; Kumar 2012 [PMID: 22521861]*

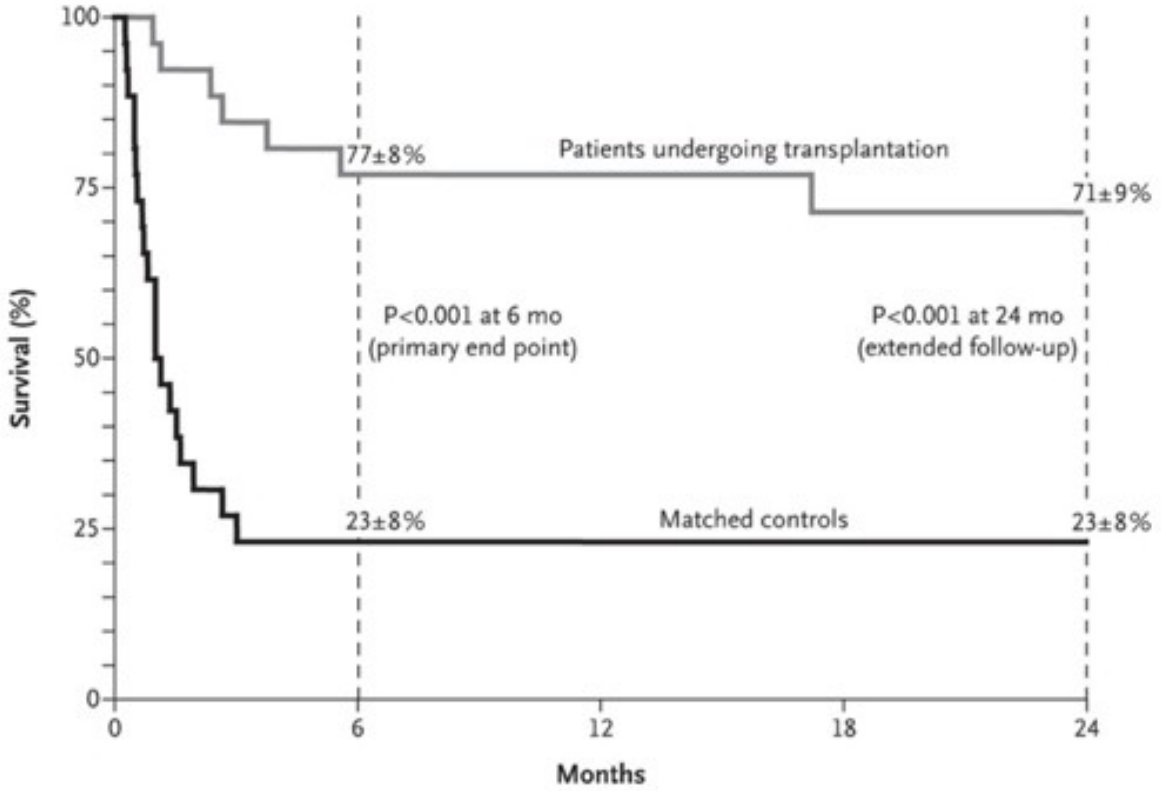
# Appendix: which ACLF?

| | EASL-CLIF | NACSELD | APASL |
|---|---|---|---|
| Who | Cirrhosis with acute decompensation | Cirrhosis, hospitalized | Chronic liver disease, with or without cirrhosis |
| Defined by | Organ failures (CLIF-C) | Two or more extrahepatic failures: shock, grade III-IV HE, dialysis, ventilation | Jaundice (bilirubin ≥ 5) and INR ≥ 1.5, then ascites or HE within 4 weeks |
| Picks up | Broader, earlier | Sicker patients | A primary liver insult |

**They find different patients. Say which one you're using**

*Kyoto consensus 2025 [PMID: 39961976]; ACG ACLF 2022 [PMID: 35006099]; AASLD 2024 [PMID: 37939273]*

# Appendix: early transplant for alcohol-associated hepatitis




| No. at Risk | | | | | |
|---|---|---|---|---|---|
| Patients undergoing transplantation | 26 | 20 | 15 | 14 | 13 |
| Matched controls | 26 | 6 | 6 | 5 | 4 |

- 26 patients with a first episode of severe alcohol-associated hepatitis, not responding to steroids
- Highly selected: first liver-related event, close family support, no severe other illness, agreed to abstinence
- 6-month survival **77% vs 23%**
- Recent drinking alone is not a reason to skip the call

*Mathurin P et al. N Engl J Med 2011 [PMID: 22070476]*

# Appendix: sources (1)


- SCCM ALF/ACLF ICU guidelines, part 1. Crit Care Med 2020 [PMID: 32058387]
- SCCM ALF/ACLF ICU guidelines, part 2. Crit Care Med 2023 [PMID: 37052436]
- EASL acute liver failure guideline. J Hepatol 2017 [PMID: 28417882]
- ACG acute liver failure guideline. Am J Gastroenterol 2023 [PMID: 37377263]
- Management of acute liver failure: update 2022. Semin Liver Dis [PMID: 36001996]
- Acute liver failure: a practical update. JHEP Rep 2024 [PMID: 39170946]
- EASL ACLF guideline. J Hepatol 2023 [PMID: 37364789]
- AASLD ACLF and critically ill cirrhosis guidance. Hepatology 2024 [PMID: 37939273], errata [PMID: 38551831], [PMID: 42467893]
- ACG ACLF guideline. Am J Gastroenterol 2022 [PMID: 35006099]
- ADQI-ICA consensus on AKI in cirrhosis. J Hepatol 2024 [PMID: 38527522]
- Surviving Sepsis Campaign 2026. Crit Care Med [PMID: 41869847]
- EASL autoimmune hepatitis guideline. J Hepatol 2025 [PMID: 40348684]
- EASL hepatic encephalopathy guideline. J Hepatol 2022 [PMID: 35724930]
- US/Canada acetaminophen poisoning consensus. JAMA Netw Open 2023 [PMID: 37552484]

# Appendix: sources (2)


- Moreau R et al. CANONIC. Gastroenterology 2013 [PMID: 23474284]
- Jalan R et al. CLIF-C ACLF score. J Hepatol 2014 [PMID: 24950482]
- Lee WM et al. NAC in non-acetaminophen ALF. Gastroenterology 2009 [PMID: 19524577]
- Clemmesen JO et al. Cerebral herniation and arterial ammonia. Hepatology 1999 [PMID: 10051463]
- Bernal W et al. Arterial ammonia, HE and intracranial hypertension. Hepatology 2007 [PMID: 17685471]
- Bhatia V et al. Arterial ammonia in ALF. Gut 2006 [PMID: 16024550]
- Kumar R et al. Persistent hyperammonemia in ALF. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]
- Cardoso FS et al. CRRT and ammonia in ALF. Hepatology 2018 [PMID: 28859230]
- Pre-emptive CRRT in ALF with cerebral edema. Aliment Pharmacol Ther 2026 [PMID: 42332991]
- Larsen FS et al. High-volume plasma exchange. J Hepatol 2016 [PMID: 26325537]
- Cavallin M et al. Terlipressin vs midodrine and octreotide. Hepatology 2015 [PMID: 25644760]
- Wong F et al. CONFIRM. NEJM 2021 [PMID: 33657294]
- Artru F et al. Transplant in ACLF grade 3. J Hepatol 2017 [PMID: 28645736]
- Mathurin P et al. Early transplant for alcoholic hepatitis. NEJM 2011 [PMID: 22070476]
- China L et al. ATTIRE. NEJM 2021 [PMID: 33657293]
- Liver transplantation for ACLF. Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]
- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- Kyoto consensus on ACLF. Hepatol Int 2025 [PMID: 39961976]
- TERLIVAZ (terlipressin) prescribing information, revised 6/2026